SIAM J. NUMER. ANAL.
Vol. 61, No. 5, pp. 2084–2105
© 2023 Society for Industrial and Applied Mathematics

AN ADAPTIVE SUPERCONVERGENT MIXED FINITE ELEMENT METHOD BASED ON LOCAL RESIDUAL MINIMIZATION*

IGNACIO MUGA†‡, SERGIO ROJAS‡, AND PATRICK VEGA§

Dedicated to the memory of Professor Ivo Babuška

Abstract. We introduce an adaptive superconvergent finite element method for a class of mixed formulations to solve partial differential equations involving a diffusion term. It combines a superconvergent postprocessing technique for the primal variable with an adaptive finite element method via residual minimization. Such a residual minimization procedure is performed on a local postprocessing scheme, commonly used in the context of mixed finite element methods. Given the local nature of that approach, the underlying saddle point problems associated with residual minimizations can be solved with minimal computational effort. We propose and study a posteriori error estimators, including the built-in residual representative associated with residual minimization schemes; and an improved estimator which adds, on the one hand, a residual term quantifying the mismatch between discrete fluxes and, on the other hand, the interelement jumps of the postprocessed solution. We present numerical experiments in two dimensions using Brezzi–Douglas–Marini elements as input for our methodology. The experiments perfectly fit our key theoretical findings and suggest that our estimates are sharp.

Key words. residual minimization, postprocessing, superconvergence, a posteriori error analysis, adaptive mesh refinement

MSC codes. 65N12, 65N15, 65N22, 65N30, 65N50

DOI. 10.1137/22M1526307

1. Introduction.

Finite element methods (FEMs) constitute one of the most popular approaches for solving boundary value problems driven by partial differential equations (PDEs). FEMs are suited to handle heterogeneous media and/or complex geometries, mainly due to the great flexibility provided by unstructured meshes. In this paper, we will focus on PDEs ruled by diffusion phenomena, where the flux $\boldsymbol{q}$ of a concentration u is described by the equation $\boldsymbol{q} = -\boldsymbol{\nabla} u$ (i.e., the flux will move from areas of higher concentration to areas of lower concentration). Mixed FEMs [13, 30] are suitable to solve such PDEs when the dual variable $\boldsymbol{q}$ is required as a fundamental unknown.[1]

*Received by the editors October 4, 2022; accepted for publication (in revised form) April 28, 2023; published electronically September 14, 2023.
https://doi.org/10.1137/22M1526307
Funding: This work was supported by the European Union's Horizon 2020 Research and Innovation Programme under Marie Sklodowska-Curie grant agreement 777778 (MATHROCKS). The first author's research was supported by the Chilean National Agency for Research and Development (ANID) through FONDECYT project 1230091, project DI Investigacion Innovadora Interdisciplinaria PUCV 039.409/2021, and project DI PUCV 039.375/2021. The second author's research was supported by Chilean grant ANID FONDECYT 3210009. The third author's research was supported by Chilean grants DI Postdoctorado 2021 (PUCV) and ANID FONDECYT 3220858.
†Basque Center for Applied Mathematics (BCAM), 48009 Bilbao, Spain.
‡Instituto de Matemáticas, Pontificia Universidad Católica de Valparaíso, 2350050 Valparaíso, Chile (ignacio.muga@pucv.cl, sergio.rojas.h@pucv.cl).
§Departamento de Matemática y Ciencia de la Computación, Universidad de Santiago de Chile, Santiago, Chile (patrick.vega@usach.cl).

[1]This is in contrast to primal-based methods, where this variable must be obtained a posteriori by differentiation, entailing a loss of accuracy [7].

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Many postprocessing techniques [3, 8, 14, 46] have been offered to obtain better approximations for the primal variable u in the context of mixed FEMs. The main idea is to enhance the approximation of u using minimal computational effort (hence, by performing local computations at the element level, which can be efficiently parallelized).

We will focus on the postprocessing technique proposed by Stenberg [46] for its conceptual simplicity. This postprocessing method (and those closely related to it) has been successfully employed with different discretization schemes to solve a wide range of PDEs, illustrating its versatility. Among them, we highlight mixed methods [35], discontinuous Galerkin (DG) methods [21], discontinuous Petrov–Galerkin (DPG) methods [31], and mimetic finite difference methods [11]. In particular, a large number of works have used the combination of hybridizable DG (HDG) methods with Stenberg's postprocessing technique on a wide variety of equations.[2]

Regarding practical implementations of FEMs, mesh (or p-) adaptivity has become an attractive strategy to limit the computational cost required for the discrete resolution of boundary value problems. Such a local adaptation procedure is commonly guided by an iterative process, driven by an a posteriori error estimator, which quantitatively *measures* the discretization error, letting practitioners decide if the accuracy of the numerical solution is enough for their applications. Consequently, the design and study of a posteriori error estimators have been of great interest in the last 40 years (see, e.g., [4, 25, 48] and the references therein). An alternative to the design of a posteriori error estimators is the use of FEMs based on residual minimization, which tend to incorporate a built-in localizable *error representative* that may be used to quantify the error and drive adaptive mesh refinements. The common procedure uses a mixed form to simultaneously solve the discrete solution and its *error representative.*

In this paper, we combine the local postprocessing idea with the concept of residual minimization to come up with a local postprocessing method that delivers, at the same time, a superconvergent postprocessed (scalar) solution and a superconvergent local error representative that is used to drive adaptive mesh refinements. The mixed system related to the residual minimization is local (i.e., element-by-element); therefore, the additional computational efforts to achieve superconvergence are negligible. We show that our postprocessed solution coincides with the one proposed by Stenberg and therefore satisfies the same a priori error estimates. Besides, we propose and study a posteriori error estimators based on the built-in residual representative associated with residual minimization schemes.

The methodology is quite general and can be applied to a class of problems exhibiting the equation $\boldsymbol{q} = -\boldsymbol{\nabla} u$ in its mixed form (see Remark 3.7). This may include the presence of convective or reactive terms. Moreover, we can incorporate variable diffusion coefficients or deal with vectorial equations (such as Stokes-like problems). The design of our estimators is (almost) independent of the underlying PDE (since it only depends on the constitutive equation $\boldsymbol{q} = -\boldsymbol{\nabla} u$) and the initial choice of finite element spaces discretizing the mixed form.

We illustrate our theoretical findings through the behavior of effectivity indices for uniform mesh refinements in the case of smooth solutions, the capability of our estimators to drive adaptive mesh refinements in the presence of singularities (recovering optimal convergence rates), and the application of our methodology to convection-diffusion problems.

[2]See, e.g., [32] for an exhaustive review of the state-of-the-art on HDG methods, most of them employing Stenberg's postprocessing.

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

1.1. A review of related literature. A posteriori error estimators employing postprocessing techniques may be found in the works [22, 33, 37, 40]. The first work in this context, due to Lovadina and Stenberg [40], is based on the a posteriori error analysis of an equivalent method, which involves the postprocessed approximation proposed in [46]. In contrast, [22, 37] plug the postprocessed approximation into the estimator without considering any equivalent scheme, while [33] used the difference between the primal discrete solution and its postprocessing to drive p-adaptivity. Moreover, in [22], the authors provide estimates based on a slight modification of the postprocessing scheme [46] in the context of HDG methods. On the other hand, [37] generalizes the work in [40] in the following sense: first, it can use any piecewise polynomial approximation of the primal variable when computing the residual; and second, it can also be applied to stabilized mixed methods.

Among other methods that include a posteriori error estimators employing postprocessing techniques, we highlight the work in mixed FEMs [45], hybrid mixed FEMs [23], and HDG methods for Kirchhoff–Love plates [34]; Stokes/Brinkman equations [5]; and Oseen equations [6], to name just a few.

In the context of FEMs based on residual minimization using the residual representative to drive adaptive refinements, we emphasize least-squares finite element methods (LS-FEM) [2, 12], DPG methods [19, 26], and the recent work on adaptive stabilized finite element methods (AS-FEMs) [18].

1.2. Outline of the paper. The remainder of the paper is organized as follows. In section 2, we introduce notation, the model problem, and a mixed finite element discretization to solve it. We also introduce Stenberg's postprocessing scheme for the mixed FEM. In section 3, we present the postprocessing based on local residual minimization and our main results, related to a posteriori error estimates. We report numerical experiments that illustrate our key findings in section 4 and draw our conclusions in section 5. We also provide appendices with some auxiliary results needed for the a posteriori theory, namely the existence of a Fortin operator and intermediate estimates needed for the proof of the reliability result.

2. Theoretical background.

2.1. Notation. Let $\Omega \subset \mathbb{R}^d$ $(d = 2, 3)$ be a polygonal/polyhedral Lipschitz domain. At the continuous level, we denote by $\mathcal{V} := L^2(\Omega)$ the space of scalar real-valued square-integrable functions defined over Ω, while its vector-valued version will be denoted by $\boldsymbol{\mathcal{V}} := \boldsymbol{L}^2(\Omega)$. Additionally, $\boldsymbol{\mathcal{H}} := H(\mathrm{div}, \Omega)$ will denote the space of functions belonging to $\boldsymbol{\mathcal{V}}$ such that its divergence belongs to $\mathcal{V}$. Besides, $H^s(\Omega) := W^{s,2}(\Omega)$ will denote the usual Sobolev space of square-summable functions up to the sth-derivative order (see [1]), $H^{1/2}(\partial\Omega)$ will denote the space of *traces* of functions in $H^1(\Omega)$, and $\langle \cdot, \cdot \rangle_{\partial\Omega}$ will denote the duality pairing between $H^{-1/2}(\partial\Omega) := (H^{1/2}(\partial\Omega))^*$ and $H^{1/2}(\partial\Omega)$. Moreover, for any $D \subset \overline{\Omega}$, we denote its corresponding Sobolev space of order s by $H^s(D)$. Additionally, we will use $(\cdot, \cdot)_D$ and $\| \cdot \|_D$ to denote the standard $L^2(D)$-inner product and inherited norm (respectively) for either scalar- or vector-valued functions.

We consider a family of meshes $\{\mathcal{T}_h\}_{h>0}$, each of them partitioning Ω into nonoverlapping simplices K, where h denotes the global mesh size defined as $h := \max_{K \in \mathcal{T}_h} h_K$, and h_K denotes the diameter of K. We assume that each mesh is conforming in the sense of [20] and shape-regular.[3] Let h_F be the diameter of a face/edge F of a simplex

[3]That is, there exists a constant $\sigma > 0$, such that $h_K/\rho_K \leq \sigma$ for all $K \in \mathcal{T}_h$ and for all $h > 0$, where ρ_K denotes the diameter of the largest ball inside K.

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

K. We further denote by $\mathcal{F}_h^{\mathrm{e}}$ the set of exterior faces/edges lying on the boundary $\partial\Omega$; by $\mathcal{F}_h^{\mathrm{i}}$ the remaining (interior) faces/edges; and by $\mathcal{F}_h := \mathcal{F}_h^{\mathrm{i}} \cup \mathcal{F}_h^{\mathrm{e}}$ the set of all faces/edges; for $F \in \mathcal{F}_h$, we define the set $\mathcal{T}_F := \{K \in \mathcal{T}_h : F \cap \partial K \neq \emptyset\}$; and for $K \in \mathcal{T}_h$, we define the sets $\mathcal{F}_K := \{F \in \mathcal{F}_h : F \subset \partial K\}$, $\mathcal{F}_K^{\mathrm{i}} := \mathcal{F}_K \cap \mathcal{F}_h^{\mathrm{i}}$, and $\mathcal{F}_K^{\mathrm{e}} := \mathcal{F}_K \cap \mathcal{F}_h^{\mathrm{e}}$.

For a scalar-valued function $w : \overline{\Omega} \to \mathbb{R}$, the jump of w across $F \in \mathcal{F}_h$ will be

$$[\![w]\!]_F := \begin{cases} w^+|_F - w^-|_F & \text{if } F = \partial K^+ \cap \partial K^- \quad (\text{with } w^\pm := w|_{K^\pm}), \\ w|_F & \text{if } F \in \mathcal{F}_h^{\mathrm{e}}, \end{cases}$$

where K^+ and K^- are such that the normal to F points from K^+ to K^-. We will omit the subscript F in $[\![\cdot]\!]_F$ if the jump is inside an $L^2(F)$-norm.

For each $K \in \mathcal{T}_h$, consider the following subspace of zero-mean functions in $H^1(K)$:

$$H^1_\star(K) := \left\{ v_K \in L^2(K) : \boldsymbol{\nabla} v_K \in \boldsymbol{L}^2(K) \text{ and } (v_K, 1)_K = 0 \right\}. \tag{2.1}$$

The space (2.1) is endowed with the norm $\|\boldsymbol{\nabla}(\cdot)\|_K$ and inner-product $\left(\boldsymbol{\nabla} u_K, \boldsymbol{\nabla} v_K\right)_K$ for all $u_K, v_K \in H^1_\star(K)$. Accordingly, we consider the following dual norm for the corresponding dual space $(H^1_\star(K))^*$ of bounded linear functionals over $H^1_\star(K)$:

$$\|\cdot\|_{(H^1_\star(K))^*} := \sup_{v_K \in H^1_\star(K)} \frac{\langle \cdot, v_K \rangle}{\|\boldsymbol{\nabla} v_K\|_K},$$

where $\langle \cdot, \cdot \rangle$ denotes the associated duality pairing.

For the broken space $H^1(\mathcal{T}_h) := \{w \in L^2(\Omega) : w|_K \in H^1(K)\ \forall K \in \mathcal{T}_h\}$, we introduce the global $H^1(\mathcal{T}_h)$-seminorm

$$\|\boldsymbol{\nabla}(\cdot)\|^2_{\mathcal{T}_h} := \sum_{K \in \mathcal{T}_h} \|\boldsymbol{\nabla}(\cdot)\|^2_K$$

and the norm $\|\cdot\|^2_{1,h} := \sum_{K \in \mathcal{T}_h} |\cdot|^2_{1,K,h}$, where

$$|\cdot|^2_{1,K,h} := \|\boldsymbol{\nabla}(\cdot)\|^2_K + \frac{1}{2} \sum_{F \in \mathcal{F}_K^{\mathrm{i}}} h_F^{-1} \left\| [\![\cdot]\!] \right\|^2_F + \sum_{F \in \mathcal{F}_K^{\mathrm{e}}} h_F^{-1} \left\| \cdot \right\|^2_F.$$

At the discrete level, for a given integer $r \geq 0$, let $\mathcal{V}_K^r$ be the space of polynomial functions over K of total degree no greater than r, and let $\boldsymbol{\mathcal{V}}_K^r := (\mathcal{V}_K^r)^d$ denote its vector-valued version. We will also consider the zero-mean space $\mathcal{V}^r_{\star,K} := \mathcal{V}_K^r \cap H^1_\star(K)$, together with the global spaces

$$\mathcal{V}_h^r := \{v_h \in L^2(\Omega) : v_h|_K \in \mathcal{V}_K^r \quad \forall K \in \mathcal{T}_h\},$$
$$\mathcal{V}^r_{\star,h} := \{v_h \in L^2(\Omega) : v_h|_K \in \mathcal{V}^r_{\star,K} \quad \forall K \in \mathcal{T}_h\}.$$

Moreover, let $\boldsymbol{\mathcal{H}}_h^r := \boldsymbol{\mathcal{V}}_h^r \cap \boldsymbol{\mathcal{H}}$, where $\boldsymbol{\mathcal{V}}_h^r := \{\boldsymbol{v}_h \in \boldsymbol{\mathcal{V}} : \boldsymbol{v}_h|_K \in \boldsymbol{\mathcal{V}}_K^r\ \forall K \in \mathcal{T}_h\}$.

Additionally, for $\boldsymbol{p} \in \boldsymbol{\mathcal{H}}$, we will also consider the norm $\|\boldsymbol{p}\|^2_{*,r} := \sum_{K \in \mathcal{T}_h} \|\boldsymbol{p}\|^2_{*,r,K}$, where $\|\cdot\|_{*,K}$ is a discrete dual norm[4] defined by

$$\|\cdot\|_{*,r,K} := \sup_{v_K \in \mathcal{V}^r_{\star,K}} \frac{(\,\cdot\, \boldsymbol{\nabla} v_K)_K}{\|\boldsymbol{\nabla} v_K\|_K}. \tag{2.2}$$

[4]Notice that $\|\cdot\|_{*,r,K}$ is a norm in the dual space of $\boldsymbol{\nabla}\mathcal{V}^r_{\star,K}$ when endowed with the vector L^2-norm, but it is not a norm in $\boldsymbol{L}^2(K)$.

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Remark 2.1. In the remainder of this paper, for a fixed integer $p \geq 1$, we will use the particular case of the discrete dual norm (2.2) and its global version with $r = p+2$. Thus, for the sake of simplicity, we will omit the polynomial degree dependence in the notation, i.e.,

$$\|\cdot\|_{*,K} := \|\cdot\|_{*,p+2,K} \qquad \text{and} \qquad \|\cdot\|_{*} := \|\cdot\|_{*,p+2}. \tag{2.3}$$

For each $K \in \mathcal{T}_h$, we denote by $Q_K^r : L^2(K) \to \mathcal{V}_K^r$ the L^2-projection from $L^2(K)$ onto $\mathcal{V}_K^r$, and $Q_h^r : L^2(\Omega) \to \mathcal{V}_h^r$ its global counterpart, namely $(Q_h^r v)|_K = Q_K^r(v|_K)$. In particular, we set $Q_h := Q_h^0$ and $Q_K := Q_K^0$ for each $K \in \mathcal{T}_h$, and we recall that the action of the operator $I - Q_K$ provides zero-mean functions on K.

In the remainder of this paper, if $A, B \geq 0$, we employ the notation $A \lesssim B$ if there exists a positive constant C, independent of h, such that $A \leq CB$.

2.2. Model problem. To fix ideas, given $f \in \mathcal{V}$ and $u_D \in H^{1/2}(\partial\Omega)$, we want to approximate the solution $u \in H^1(\Omega)$ and the flux $\boldsymbol{q} := -\boldsymbol{\nabla} u \in \boldsymbol{\mathcal{V}}$ of the model problem

$$\left\{ \begin{aligned} -\Delta u &= f && \text{in } \Omega, \\ u &= u_D && \text{on } \partial\Omega, \end{aligned} \right. \iff \left\{ \begin{aligned} \boldsymbol{q} + \boldsymbol{\nabla} u &= \boldsymbol{0} && \text{in } \Omega, \\ \boldsymbol{\nabla} \cdot \boldsymbol{q} &= f && \text{in } \Omega, \\ u &= u_D && \text{on } \partial\Omega, \end{aligned} \right. \tag{2.4}$$

which admits the following variational formulation: Find $(u, \boldsymbol{q}) \in \mathcal{V} \times \boldsymbol{\mathcal{H}}$ such that

$$(\boldsymbol{q}, \boldsymbol{p})_\Omega - (\boldsymbol{\nabla} \cdot \boldsymbol{p}, u)_\Omega = -\langle u_D, \boldsymbol{p} \cdot \boldsymbol{n} \rangle_{\partial\Omega} \qquad \forall \boldsymbol{p} \in \boldsymbol{\mathcal{H}}, \tag{2.5a}$$

$$(\boldsymbol{\nabla} \cdot \boldsymbol{q}, v)_\Omega = (f, v)_\Omega \qquad \forall v \in \mathcal{V}. \tag{2.5b}$$

Formulation (2.5) is well-posed by means of the Babuška–Brezzi[5] theory (see, e.g., [30, Theorem 49.3]).

2.3. A mixed FEM. As an illustrative example of a mixed FEM discretizing (2.5), we present the d-dimensional Brezzi–Douglas–Marini (BDM) finite element family [16, 17], together with its a priori error estimates. The well-posedness of this discrete formulation relies on the standard Babuška–Brezzi theory (see, e.g., [30, section 50.1.1]).

For a fixed integer $p \geq 1$, we consider the following conforming and fully discrete version of (2.5): Find $(u_h, \boldsymbol{q}_h) \in \mathcal{V}_h^{p-1} \times \boldsymbol{\mathcal{H}}_h^p$ such that

$$(\boldsymbol{q}_h, \boldsymbol{p}_h)_{\mathcal{T}_h} - (\boldsymbol{\nabla} \cdot \boldsymbol{p}_h, u_h)_{\mathcal{T}_h} = -\langle u_D, \boldsymbol{p}_h \cdot \boldsymbol{n} \rangle_{\partial\Omega} \qquad \forall \boldsymbol{p}_h \in \boldsymbol{\mathcal{H}}_h^p \subset \boldsymbol{\mathcal{H}}, \tag{2.6a}$$

$$(\boldsymbol{\nabla} \cdot \boldsymbol{q}_h, v_h)_{\mathcal{T}_h} = (f, v_h)_{\mathcal{T}_h} \qquad \forall v_h \in \mathcal{V}_h^{p-1} \subset \mathcal{V}. \tag{2.6b}$$

Now, we recall the a priori error estimates for (2.6); cf. [46, Theorem 2.1].

THEOREM 2.2. *Let $(u, \boldsymbol{q}) \in \mathcal{V} \times \boldsymbol{\mathcal{H}}$ solve (2.5) and $(u_h, \boldsymbol{q}_h) \in \mathcal{V}_h^{p-1} \times \boldsymbol{\mathcal{H}}_h^p$ solve (2.6). Let us suppose that u lies in $H^s(\Omega)$ with $s > 1$. Then, we have*

$$\|\boldsymbol{q} - \boldsymbol{q}_h\|_\Omega \lesssim h^r |\boldsymbol{q}|_{r,\Omega}, \qquad \text{with } r = \min\{s-1, p+1\}, \tag{2.7a}$$

$$\|u - u_h\|_\Omega \lesssim h^l (|\boldsymbol{q}|_{l,\Omega} + |u|_{l,\Omega}), \qquad \text{with } l = \min\{s-1, p\}, \tag{2.7b}$$

[5]The Babuška–Brezzi theory, based on the pioneering works of Ivo Babuška (1926–2023) in 1971 [9] and Franco Brezzi in 1974 [15], establishes stable mixed formulations for the numerical approximation of PDEs using FEMs, providing a fundamental tool in numerical analysis.

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

where $|\cdot|_{s,\Omega}$ *is the standard* $H^s(\Omega)$*-seminorm* [1]. *If* Ω *is convex, then we have*

$$\|u-u_h\|_\Omega \lesssim h^l(|\boldsymbol{q}|_{l-1,\Omega}+|u|_{l,\Omega}), \quad \text{with } l=\min\{s,p\}, \tag{2.8}$$

$$\|u_h-Q_h^{p-1}u\|_\Omega \lesssim \begin{cases} h^{r+1}|\boldsymbol{q}|_{r,\Omega}, & \text{with } r=\min\{s-1,p+1\} \quad \text{for } p\geq 2,\\ h^2|\boldsymbol{q}|_{2,\Omega} & \text{for } p=1. \end{cases} \tag{2.9}$$

If $f\in\mathcal{V}_h^{p-1}$, then the first estimate in (2.9) also holds for $p=1$.

Remark 2.3. Since $\|\boldsymbol{q}-\boldsymbol{q}_h\|_{*,K}\leq\|\boldsymbol{q}-\boldsymbol{q}_h\|_K$, for all $K\in\mathcal{T}_h$, we have the same a priori error estimate (2.7a) for $\|\boldsymbol{q}-\boldsymbol{q}_h\|_*$ (see Remark 2.1 for the definition of the norms).

2.4. A superconvergent postprocessing. Let $(u_h,\boldsymbol{q}_h)$ solve (2.6). We present a slight modification (cf. [40, equations (2.32) and (2.33)]) of the postprocessing scheme introduced by Stenberg in [46].[6] We look for a discrete solution $\widetilde{u}_h\in\mathcal{V}_h^{p+1}$ satisfying the following two equations on each element $K\in\mathcal{T}_h$:

$$(\boldsymbol{\nabla}\widetilde{u}_h,\boldsymbol{\nabla}v_K)_K=-(\boldsymbol{q}_h,\boldsymbol{\nabla}v_K)_K \qquad \forall\, v_K\in\mathcal{V}_{\star,K}^{p+1}, \tag{2.10a}$$

$$(\widetilde{u}_h,1)_K=(u_h,1)_K. \tag{2.10b}$$

Remark 2.4. Since $\widetilde{u}_h$ is computed locally, the computational effort required to obtain $\widetilde{u}_h$ is negligible compared to the computational effort needed to solve the global problem (2.6).

For the postprocessed solution $\widetilde{u}_h$, the next theorem shows an a priori error estimate (cf. [46, Theorem 2.2]). We recall its proof for the sake of completeness and to emphasize its dependence on a priori error estimates associated with the mixed method (2.6).

THEOREM 2.5. *Let* $(u,\boldsymbol{q})\in\mathcal{V}\times\boldsymbol{\mathcal{H}}$ *solve* (2.5), *and let* $\widetilde{u}_h$ *be defined by* (2.10). *Then, for all* $K\in\mathcal{T}_h$, *we have*

$$\begin{aligned}\|u-\widetilde{u}_h\|_K\leq\frac{h_K}{\pi}\Big(\|\boldsymbol{q}-\boldsymbol{q}_h\|_K+\Big\|\boldsymbol{\nabla}\big(u-Q_K^{p+1}u|_K\big)\Big\|_K\Big)\\ +\Big\|u_h-Q_K^{p-1}u|_K\Big\|_K+\Big\|u-Q_K^{p+1}u|_K\Big\|_K.\end{aligned} \tag{2.11}$$

Proof. Let us define $v\in\mathcal{V}_{\star,h}^{p+1}$ through $v|_K=(I-Q_K)\big(Q_K^{p+1}u|_K-\widetilde{u}_h|_K\big)$ for each $K\in\mathcal{T}_h$. Thanks to (2.10a) and the fact that $\boldsymbol{\nabla}u+\boldsymbol{q}=0$, we write

$$\begin{aligned}\|\boldsymbol{\nabla}v\|_K^2&=\Big(\boldsymbol{\nabla}(I-Q_K)\big(Q_K^{p+1}u|_K-\widetilde{u}_h|_K\big),\boldsymbol{\nabla}v\Big)_K\\ &=\Big(\boldsymbol{\nabla}\big(Q_K^{p+1}u|_K-\widetilde{u}_h\big),\boldsymbol{\nabla}v\Big)_K\\ &=\Big(\boldsymbol{\nabla}Q_K^{p+1}u|_K,\boldsymbol{\nabla}v\Big)_K+(\boldsymbol{q}_h,\boldsymbol{\nabla}v)_K\\ &=-\Big(\boldsymbol{\nabla}\big(u-Q_K^{p+1}u|_K\big),\boldsymbol{\nabla}v\Big)_K-(\boldsymbol{q}-\boldsymbol{q}_h,\boldsymbol{\nabla}v)_K.\end{aligned}$$

Thus, by using the Cauchy–Schwarz inequality, we obtain

$$\|\boldsymbol{\nabla}v\|_K\leq\Big\|\boldsymbol{\nabla}\big(u-Q_K^{p+1}u|_K\big)\Big\|_K+\|\boldsymbol{q}-\boldsymbol{q}_h\|_K. \tag{2.12}$$

[6] In [46], the author uses $(f,v_K)_K-\langle\boldsymbol{q}_h\cdot\boldsymbol{n},v_K\rangle_{\partial K}$ as the right-hand side of (2.10a).

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Since $v|_K \in \mathcal{V}_{\star,K}^{p+1}$, thanks to (2.12) and Poincaré inequality [43, equation (4.3)], we have

$$\|v\|_K \leq \frac{h_K}{\pi}\|\boldsymbol{\nabla} v\|_K \leq \frac{h_K}{\pi}\left(\left\|\boldsymbol{\nabla}\left(u - Q_K^{p+1}u|_K\right)\right\|_K + \|\boldsymbol{q} - \boldsymbol{q}_h\|_K\right). \tag{2.13}$$

On the other hand, the definitions of the orthogonal projections Q_K, Q_K^{p-1}, and Q_K^{p+1} let us deduce that $Q_K Q_K^{p+1} u|_K = Q_K Q_K^{p-1} u|_K$. Moreover, recalling (2.10b) and the boundedness of Q_K, we conclude that

$$\left\|Q_K\left(Q_K^{p+1}u|_K - \widetilde{u}_h|_K\right)\right\|_K = \left\|Q_K\left(Q_K^{p-1}u|_K - u_h|_K\right)\right\|_K \leq \left\|Q_K^{p-1}u|_K - u_h\right\|_K. \tag{2.14}$$

Hence, (2.11) follows from (2.13), (2.14), and the splitting (for all $K \in \mathcal{T}_h$)

$$(u - \widetilde{u}_h)|_K = v|_K + Q_K\left(Q_K^{p+1}u|_K - \widetilde{u}_h|_K\right) + \left(u|_K - Q_K^{p+1}u|_K\right). \qquad \square$$

The extension of this analysis to the other mixed methods in [16, 17, 41, 44, 47] is straightforward.

Remark 2.6 (optimal convergence). Thanks to the approximation properties of Q_h^{p+1} and the estimates of Theorems 2.2 and 2.5, if $u \in H^s(\Omega)$, with $s > 1$, and Ω is convex, then we have

$$\|u - \widetilde{u}_h\|_{\mathcal{T}_h} \lesssim \begin{cases} h^{r+1}(|\boldsymbol{q}|_{r,\Omega} + |u|_{r+1,\Omega}), & r = \min\{s-1, p+1\} \quad \text{for } p \geq 2, \\ h^2(|\boldsymbol{q}|_{2,\Omega} + |u|_{2,\Omega}) & \text{for } p = 1 \end{cases} \tag{2.15}$$

and

$$\|\boldsymbol{\nabla}(u - \widetilde{u}_h)\|_{\mathcal{T}_h} \lesssim h^r(|\boldsymbol{q}|_{r,\Omega} + |u|_{r+1,\Omega}), \quad r = \min\{s-1, p+1\} \quad \text{for } p \geq 1. \tag{2.16}$$

Moreover, if $f \in \mathcal{V}_h^{p-1}$, then the first estimate in (2.15) also holds for $p = 1$.

3. Main results. In this section, we introduce our postprocessing scheme and state its relation with the postprocessing scheme proposed by Stenberg [46]. Next, we present our main results concerning a posteriori error estimates.

3.1. Postprocessing based on local residual minimization. We present the main idea behind our postprocessing procedure, which considers the local postprocessing scheme (2.10) as a starting point.

In our case, by keeping the same approximation space $\mathcal{V}_h^{p+1}$, we will follow a residual minimization approach. Consequently, the new postprocessed variable $\nu_h \in \mathcal{V}_h^{p+1}$ will be defined locally through the following set of local residual minimization problems: Find $\nu_h \in \mathcal{V}_h^{p+1}$ such that for each $K \in \mathcal{T}_h$,

$$\nu_h|_K = \underset{w_K \in \mathcal{V}_K^{p+1}}{\arg\min} \frac{1}{2}\|\boldsymbol{q}_h + \boldsymbol{\nabla} w_K\|_{*,K}^2, \tag{3.1a}$$

$$(\nu_h, 1)_K = (u_h, 1)_K, \tag{3.1b}$$

where $\|\cdot\|_{*,K}$ is the discrete dual norm defined in (2.3).

For each $K \in \mathcal{T}_h$, let $\nu_K := \nu_h|_K$. We notice that solving (3.1) is equivalent to solving the local mixed linear system (see [24]): Find $(\varepsilon_K, \nu_K) \in \mathcal{V}_{\star,K}^{p+2} \times \mathcal{V}_K^{p+1}$ such that

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

$$(\boldsymbol{\nabla}\varepsilon_K, \boldsymbol{\nabla} v_K)_K + (\boldsymbol{\nabla}\nu_K, \boldsymbol{\nabla} v_K)_K = -(\boldsymbol{q}_h, \boldsymbol{\nabla} v_K)_K \qquad \forall v_K \in \mathcal{V}_{\star,K}^{p+2}, \tag{3.2a}$$

$$(\boldsymbol{\nabla} w_K, \boldsymbol{\nabla}\varepsilon_K)_K = 0 \qquad \forall w_K \in \mathcal{V}_{\star,K}^{p+1}, \tag{3.2b}$$

$$(\nu_h, 1)_K = (u_h, 1)_K. \tag{3.2c}$$

We also introduce $\varepsilon_h \in \mathcal{V}_{\star,h}^{p+2}$ as the global version of ε_K; that is, $\varepsilon_h|_K := \varepsilon_K$ for each $K \in \mathcal{T}_h$. Next, we establish how our postprocessed solution ν_h relates to the Stenberg postprocess. In particular, both postprocessed solutions, (2.10) and (3.1), are the same. However, they have a different nature: the first is constructed through a residual minimization approach, while the second directly solves an elliptic problem.

PROPOSITION 3.1. *Let $\widetilde{u}_h \in \mathcal{V}_h^{p+1}$ solve the postprocessing problem* (2.10)*, and let $\nu_h \in \mathcal{V}_h^{p+1}$ solve the residual minimization postprocessing problem* (3.1)*. Then, $\nu_h = \widetilde{u}_h$. That is, both postprocessed solutions are the same.*

Proof. Since $\mathcal{V}_{\star,K}^{p+1} \subset \mathcal{V}_{\star,K}^{p+2}$, for each $K \in \mathcal{T}_h$, we can test (3.2a) with $v_K \in \mathcal{V}_{\star,K}^{p+1}$. Moreover, thanks to (3.2b), the first term in (3.2a) vanishes, so we have that ν_K solves (2.10). By uniqueness of the solution of (2.10), we conclude that $\nu_h = \widetilde{u}_h$. ∎

Given the fact that both postprocessed solutions are equal, hereafter we denote them uniquely by ν_h.

COROLLARY 3.2 (a priori error estimate). *The postprocessed solution $\nu_h \in \mathcal{V}_h^{p+1}$ also satisfies the a priori error estimate* (2.11) *given in Theorem* 2.5.

Remark 3.3 (general test spaces). We could have considered $\mathcal{V}_{\star,K} \subset H_\star^1(K)$ as an arbitrary finite-dimensional test space containing $\mathcal{V}_{\star,K}^{p+1}$ in (3.2a). In such a case, Proposition 3.1 and Corollary 3.2 still hold. However, we prefer $\mathcal{V}_{\star,K}^{p+2}$ since higher-order polynomials will not increase the order of convergence.

Remark 3.4. The local residual representative $\varepsilon_K \in \mathcal{V}_{\star,K}^{p+2}$ is nothing but the Riesz representative of the local residual $-(\boldsymbol{q}_h + \boldsymbol{\nabla}\nu_h, \boldsymbol{\nabla}(\cdot))_K \in (\mathcal{V}_{\star,K}^{p+2})^*$. In general, $\varepsilon_K \neq 0$ since $\mathcal{V}_{\star,K}^{p+2}$ strictly contains $\mathcal{V}_{\star,K}^{p+1}$. Hence, $\|\boldsymbol{\nabla}\varepsilon_K\|_K$ can be used as a local (a posteriori) error estimator to drive adaptivity.

3.2. A posteriori error analysis. In this section, we introduce and analyze a posteriori error estimators based on the local residual representative ε_K provided by (3.2). In contrast to DPG methods, which directly use the (built-in) residual representative as an (a posteriori) error estimator, we further propose an improved error estimator using ε_K as a starting point (see section 3.2.2).

3.2.1. A built-in a posteriori error estimator. In this section, we study the a posteriori error estimator $\widetilde{\eta} := \left(\sum_{K\in\mathcal{T}_h} \widetilde{\eta}_K^2\right)^{1/2}$, with $\widetilde{\eta}_K := \|\boldsymbol{\nabla}\varepsilon_K\|_K$, where $\varepsilon_K \in \mathcal{V}_{\star,K}^{p+2}$ is the first component of the solution of problem (3.2). We have realized that the associated reliability estimate allows controlling the error $\boldsymbol{q} - \boldsymbol{q}_h$, only in a norm weaker than the $L^2(\Omega)$-norm. As in [40], full control of the $L^2(\Omega)$-norm requires additional terms quantifying the mismatch associated with the discrete version of the equation $\boldsymbol{q} + \boldsymbol{\nabla} u = \boldsymbol{0}$ (namely, $\|\boldsymbol{q}_h + \boldsymbol{\nabla}\nu_h\|_K$), together with interelement jumps of the postprocessed solution ν_h (see section 3.2.2 for the details).

To state the last assumption needed to prove the reliability estimate, we introduce the following auxiliary problem: Find $\theta_h \in \mathcal{V}_h^{p+2}$ such that, for each $K \in \mathcal{T}_h$,

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

$$(\boldsymbol{\nabla}\theta_h, \boldsymbol{\nabla}v_K)_K = -(\boldsymbol{q}_K, \boldsymbol{\nabla}v_K)_K \qquad \forall v_K \in \mathcal{V}_{\star,K}^{p+2}, \tag{3.3a}$$

$$(\theta_h, 1)_K = (u_h, 1)_K. \tag{3.3b}$$

Now, we introduce the last assumption.

Assumption 3.5 (saturation). Let $(\varepsilon_h, \nu_h) \in \mathcal{V}_{\star,h}^{p+2} \times \mathcal{V}_h^{p+1}$ solve (3.2) and $\theta_h \in \mathcal{V}_h^{p+2}$ solve (3.3). There exists an h-independent real number $\delta \in [0,1)$ such that

$$\|\boldsymbol{\nabla}(u-\theta_h)\|_{\mathcal{T}_h} \leq \delta \|\boldsymbol{\nabla}(u-\nu_h)\|_{\mathcal{T}_h}.$$

THEOREM 3.6 (reliability based on the residual representative). *Let $(u, \boldsymbol{q}) \in \mathcal{V}\times\boldsymbol{\mathcal{H}}$ solve the continuous problem* (2.5)*; let $(u_h, \boldsymbol{q}_h) \in \mathcal{V}_h^{p-1}\times\boldsymbol{\mathcal{H}}_h^p$ solve the discrete problem* (2.6)*; and let $(\varepsilon_h, \nu_h) \in \mathcal{V}_{\star,h}^{p+2} \times \mathcal{V}_h^{p+1}$ solve* (3.2)*. If Assumption* 3.5 *is satisfied, then the following estimate holds:*

$$\|\boldsymbol{\nabla}(u-\nu_h)\|_{\mathcal{T}_h} + \|\boldsymbol{q}-\boldsymbol{q}_h\|_* \lesssim \widetilde{\eta}. \tag{3.4}$$

Proof. The proof requires auxiliary results that have been shifted to Appendix A. Let θ_h solve (3.3). Thanks to (A.2) and after summing over all $K \in \mathcal{T}_h$, we get

$$\|\boldsymbol{\nabla}(\theta_h-\nu_h)\|_{\mathcal{T}_h} = \widetilde{\eta}. \tag{3.5}$$

Combining (3.5) with Assumption 3.5 and the triangle inequality, we obtain

$$\|\boldsymbol{\nabla}(u-\nu_h)\|_{\mathcal{T}_h} \leq \frac{1}{1-\delta}\widetilde{\eta}. \tag{3.6}$$

The result follows from (3.6) and adding (A.3) over all $K \in \mathcal{T}_h$. ∎

Remark 3.7 (PDE independence). Notice that our a posteriori estimates can be applied to more general PDEs (with suitable boundary conditions) of the form

$$(-\Delta + \mathcal{R})u = f \quad \Longleftrightarrow \quad \begin{cases} \boldsymbol{q} + \boldsymbol{\nabla}u &= 0, \\ \boldsymbol{\nabla}\cdot\boldsymbol{q} + \mathcal{R}u &= f, \end{cases} \tag{3.7}$$

where $\mathcal{R}$ is a nondiffusive differential operator, and $(u, \boldsymbol{q}) \in H^1(\Omega)\times H(\mathrm{div}, \Omega)$ satisfies $\mathcal{R}u \in L^2(\Omega)$. Enjoying such a property was one of the main reasons why we have used the discrete dual norm $\|\cdot\|_*$. In fact, if we had proceeded to bound $\|\boldsymbol{q}-\boldsymbol{q}_h\|_\Omega$ as in [40], we would have found the quantity $\boldsymbol{\nabla}\cdot(\boldsymbol{q}-\boldsymbol{q}_h)$ (see [40, Theorems 3.2 and 3.3]), which generate PDE-dependent additional terms in the error estimates since

$$\boldsymbol{\nabla}\cdot(\boldsymbol{q}-\boldsymbol{q}_h) = f - \boldsymbol{\nabla}\cdot\boldsymbol{q}_h - \mathcal{R}\nu_h - \mathcal{R}(u-\nu_h).$$

To properly bound $\|\boldsymbol{q}-\boldsymbol{q}_h\|_\Omega$ without losing PDE independence, we will include additional terms in an improved a posteriori estimator (see section 3.2.2).

To close this section, we state and prove our local efficiency estimate.

THEOREM 3.8 (local efficiency). *The following holds true for all $K \in \mathcal{T}_h$:*

$$\widetilde{\eta}_K \leq \|\boldsymbol{\nabla}(u-\nu_h)\|_K + \|\boldsymbol{q}-\boldsymbol{q}_h\|_{*,K}. \tag{3.8}$$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Proof. Thanks to (3.2a) and the fact that $\boldsymbol{q}+\boldsymbol{\nabla} u=0$, for each $K\in\mathcal{T}_h$, we have

$$\begin{aligned}(\boldsymbol{\nabla}\varepsilon_K,\boldsymbol{\nabla} v_K)_K&=-(\boldsymbol{\nabla}\nu_K,\boldsymbol{\nabla} v_K)_K-(\boldsymbol{q}_h,\boldsymbol{\nabla} v_K)_K\\&=(\boldsymbol{\nabla}(u-\nu_K),\boldsymbol{\nabla} v_K)_K+(\boldsymbol{q}-\boldsymbol{q}_h,\boldsymbol{\nabla} v_K)_K\end{aligned}$$

for all $v_K\in\mathcal{V}_{\star,K}^{p+2}$. Thus, (3.8) follows since

$$\widetilde{\eta}_K=\sup_{v_K\in\mathcal{V}_{\star,K}^{p+2}}\frac{(\boldsymbol{\nabla}\varepsilon_K,\boldsymbol{\nabla} v_K)_K}{\|\boldsymbol{\nabla} v_K\|_K}\le\|\boldsymbol{\nabla}(u-\nu_K)\|_K+\|\boldsymbol{q}-\boldsymbol{q}_h\|_{*,K}.$$ ∎

3.2.2. An improved a posteriori error estimator. We study here the behavior of an alternative to the estimator $\widetilde{\eta}_K:=\|\boldsymbol{\nabla}\varepsilon_K\|_K$, which lets us deal with estimates related to $\|\boldsymbol{q}-\boldsymbol{q}\|_K$ (instead of the discrete dual norm $\|\boldsymbol{q}-\boldsymbol{q}_h\|_{*,K}$). Let us define the estimator $\eta:=\big(\sum_{K\in\mathcal{T}_h}\eta_K^2\big)^{1/2}$, with

$$\eta_K^2:=\widetilde{\eta}_K^2+\|\boldsymbol{q}_h+\boldsymbol{\nabla}\nu_h\|_K^2+\frac{1}{2}\sum_{F\in\mathcal{F}_K^{\mathrm{i}}}h_F^{-1}\|[\![\nu_h]\!]\|_F^2+\sum_{F\in\mathcal{F}_K^{\mathrm{e}}}h_F^{-1}\|u_D-\nu_h\|_F^2.\tag{3.9}$$

To start, we present the reliability result for our improved estimator η.

THEOREM 3.9 (reliability, improved estimator). *Let $(u,\boldsymbol{q})\in\mathcal{V}\times\boldsymbol{\mathcal{H}}$ solve the continuous problem* (2.5)*; let $(u_h,\boldsymbol{q}_h)\in\mathcal{V}_h^{p-1}\times\boldsymbol{\mathcal{H}}_h^p$ solve the discrete problem* (2.6)*; and let $(\varepsilon_h,\nu_h)\in\mathcal{V}_{\star,h}^{p+2}\times\mathcal{V}_h^{p+1}$ solve* (3.2)*. If Assumption* 3.5 *is satisfied, then the following estimate holds:*

$$\|u-\nu_h\|_{1,h}+\|\boldsymbol{q}-\boldsymbol{q}_h\|_\Omega\lesssim\eta.\tag{3.10}$$

Proof. Thanks to (A.2) and the fact that $\widetilde{\eta}_K\le\eta_K$, for each $K\in\mathcal{T}_h$, we have

$$\|\boldsymbol{\nabla}(\theta_h-\nu_h)\|_K\le\eta_K.\tag{3.11}$$

Additionally, since $\|\boldsymbol{q}-\boldsymbol{q}_h\|_K\le\|\boldsymbol{q}_h+\boldsymbol{\nabla}\nu_h\|_K+\|\boldsymbol{\nabla}(u-\nu_h)\|_K$, we conclude that

$$\|\boldsymbol{q}-\boldsymbol{q}_h\|_K\le\eta_K+\|\boldsymbol{\nabla}(u-\nu_h)\|_K.\tag{3.12}$$

Since $u\in H^1(\Omega)$ and $u=u_D$ on $\partial\Omega$, for $F_{\mathrm{i}}\in\mathcal{F}_K^{\mathrm{i}}$ and $F_{\mathrm{e}}\in\mathcal{F}_K^{\mathrm{e}}$, we have

$$\big\|[\![u-\nu_h]\!]\big\|_{F_{\mathrm{i}}}=\big\|[\![\nu_h]\!]\big\|_{F_{\mathrm{i}}}\qquad\text{and}\qquad\|u-\nu_h\|_{F_{\mathrm{e}}}=\|u_D-\nu_h\|_{F_{\mathrm{e}}}.\tag{3.13}$$

Finally, (3.10) is a direct consequence of adding (3.11)–(3.13) over all $K\in\mathcal{T}_h$ and Assumption 3.5 (to get an upper bound for $\|\boldsymbol{\nabla}(u-\nu_h)\|_{\mathcal{T}_h}$, as in Theorem 3.6). ∎

Remark 3.10 (control of $\|u-\nu_h\|_\Omega$). We can also get an upper bound for the error $\|u-\nu_h\|_\Omega$ through the use of a smoothing linear operator, such as the Oswald interpolation operator $\mathcal{I}_{\mathrm{av}}^{\mathcal{T}_h}:\mathcal{V}_h^{p+1}\to\mathcal{V}_h^{p+1}\cap H^1(\Omega)$ in the context of diffusive PDEs (see [27, section 5.2.2]). Indeed, thanks to the Poincaré inequality, the approximation properties of $\mathcal{I}_{\mathrm{av}}^{\mathcal{T}_h}$ [36, Theorem 2.1], Theorem 3.9, and the definition of η, we conclude that

$$\begin{aligned}\|u-\nu_h\|_\Omega&\le\|u-\mathcal{I}_{\mathrm{av}}^{\mathcal{T}_h}\nu_h\|_\Omega+\|\nu_h-\mathcal{I}_{\mathrm{av}}^{\mathcal{T}_h}\nu_h\|_\Omega\\&\lesssim\|\boldsymbol{\nabla}(u-\mathcal{I}_{\mathrm{av}}^{\mathcal{T}_h}\nu_h)\|_{\mathcal{T}_h}+\sum_{F\in\mathcal{F}_h}h_F^{1/2}\|[\![\nu_h]\!]\|_F\end{aligned}$$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

$$\begin{aligned}
&\leq \|\boldsymbol{\nabla}(u-\nu_h)\|_{\mathcal{T}_h} + \|\boldsymbol{\nabla}(\nu_h - \mathcal{I}_{\mathrm{av}}^{\mathcal{T}_h}\nu_h)\|_{\mathcal{T}_h} + \sum_{F\in\mathcal{F}_h} h_F^{1/2}\|[\![\nu_h]\!]\|_F \\
&\lesssim \|\boldsymbol{\nabla}(u-\nu_h)\|_{\mathcal{T}_h} + \sum_{F\in\mathcal{F}_h} h_F^{-1/2}\|[\![\nu_h]\!]\|_F + \sum_{F\in\mathcal{F}_h} h_F^{1/2}\|[\![\nu_h]\!]\|_F \\
&\lesssim \eta,
\end{aligned}$$

assuming that $h_F \leq 1$ for all $F \in \mathcal{F}_h$.

To conclude this section, we derive the following local efficiency estimate.

THEOREM 3.11 (local efficiency, improved estimator). *For all $K \in \mathcal{T}_h$, the following estimate holds true:*

$$\eta_K \leq |u-\nu_h|_{1,K,h} + \|\boldsymbol{q}-\boldsymbol{q}_h\|_K\,. \tag{3.14}$$

Proof. Notice the identities

$$\big\|[\![\nu_h]\!]\big\|_{F_{\mathrm{i}}} = \big\|[\![u-\nu_h]\!]\big\|_{F_{\mathrm{i}}} \qquad \text{and} \qquad \|u_D-\nu_h\|_{F_{\mathrm{e}}} = \|u-\nu_h\|_{F_{\mathrm{e}}}$$

for $F_{\mathrm{i}} \in \mathcal{F}_K^{\mathrm{i}}$ and $F_{\mathrm{e}} \in \mathcal{F}_K^{\mathrm{e}}$, respectively. Thus, (3.14) follows from (3.8), the triangle inequality $\|\boldsymbol{q}_h + \boldsymbol{\nabla}\nu_h\|_K \leq \|\boldsymbol{\nabla}(u-\nu_h)\|_K + \|\boldsymbol{q}-\boldsymbol{q}_h\|_K$, and the fact that $\|\boldsymbol{q}-\boldsymbol{q}_h\|_{*,K} \leq \|\boldsymbol{q}-\boldsymbol{q}_h\|_K$ for all $K \in \mathcal{T}_h$. ∎

Remark 3.12 (control of $h_K^{1/2}\|(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}\|_{\partial K}$). We can also include the standard term $h_K^{1/2}\|(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}\|_{\partial K}$ as part of the error in the a posteriori error estimates. However, to control $h_K^{1/2}\|(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}\|_{\partial K}$, we had to include an extra (PDE-dependent) additive term in our improved estimator η_K, namely

$$h_K\|f - \boldsymbol{\nabla}\cdot\boldsymbol{q}_h - \mathcal{R}\nu_h\|_K, \tag{3.15}$$

where $\mathcal{R}$ is the nondiffusive counterpart of the underlying differential operator (see Remark 3.7). To obtain reliability and efficiency results for such a modification of η_K (which would be the analogues of Theorems 3.9 and 3.11), we require lower and upper bounds for the term (3.15). In particular, to get such bounds, the flux error should also consider the contribution of $h_K\|\boldsymbol{\nabla}\cdot(\boldsymbol{q}-\boldsymbol{q}_h)\|_K$ (see Appendix B for details). Our proof assumes the existence of a particular Fortin operator (Assumption B.1), and Appendix C is devoted to verifying that assumption in the lowest-order case.

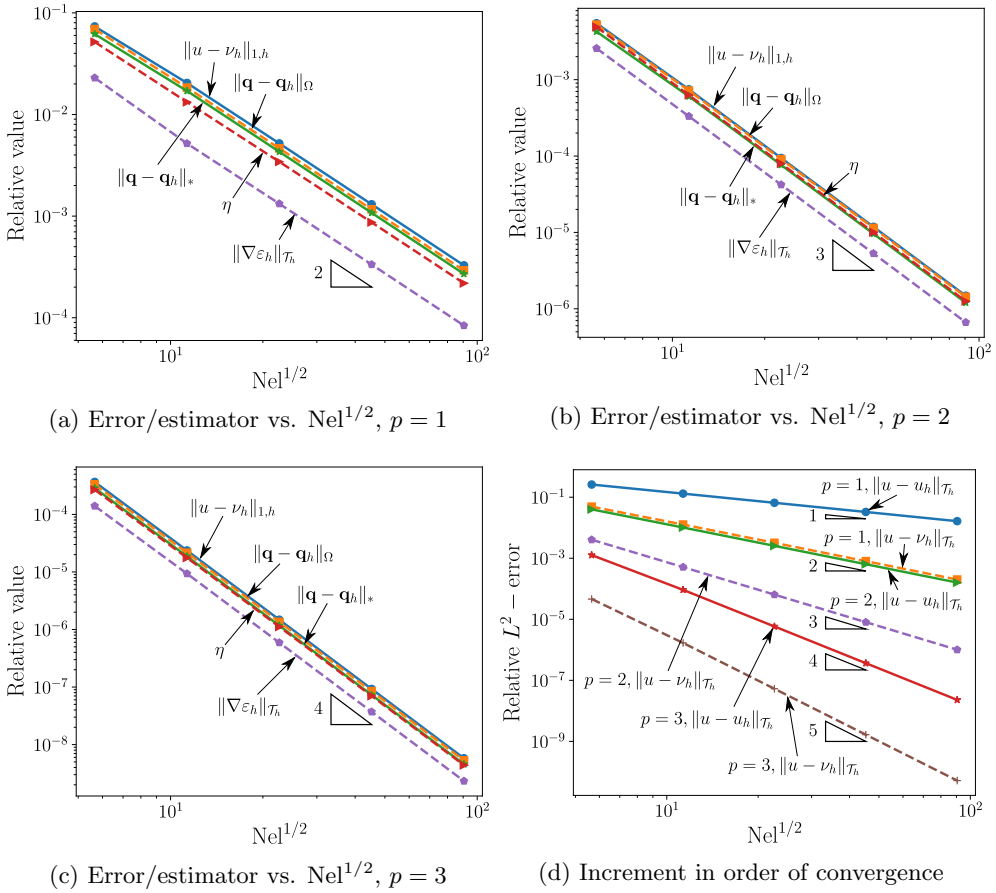
4. Numerical validation. In this section, we present three numerical examples (in two dimensions) to illustrate the performance of our scheme and validate our theoretical findings. We take a BDM mixed finite element discretization [16, 17] as the input for our scheme. We consider h-adaptive iterative refinements combining Dörfler marking [28] together with a bisection-type refinement criterion [10]. Our computational implementation is based on the open-source finite element libraries Fenics [38] and Dolphin [39].

4.1. A smooth solution. We consider the Poisson problem (2.4) over the unit square $\Omega := (0,1)^2 \subset \mathbb{R}^2$. We set $u_D(x_1,x_2) := 0$ on $\partial\Omega$ and f such that the exact solution is given by $u(x_1,x_2) := x_1(1-x_1)\sin(\pi x_2)$ in Ω. We study the behavior of our scheme together with our two a posteriori error estimators. For uniformly refined meshes, we analyze convergence rates and the effectivity index η/error, where the error expression is given by $(\|u-\nu_h\|_{1,h}^2 + \|\boldsymbol{q}-\boldsymbol{q}_h\|_\Omega^2)^{1/2}$.

For different polynomial orders p, Figures 1(a), 1(b), and 1(c) show the convergence curves associated with each error measurement, namely $\|u-\nu_h\|_{1,h}$, $\|\boldsymbol{q}-\boldsymbol{q}_h\|_\Omega$,

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy



(a) Error/estimator vs. Nel$^{1/2}$, $p=1$

(b) Error/estimator vs. Nel$^{1/2}$, $p=2$

(c) Error/estimator vs. Nel$^{1/2}$, $p=3$

(d) Increment in order of convergence

FIG. 1. *Convergence rates for a smooth solution scenario, uniform mesh refinements.*

and $\|\boldsymbol{q}-\boldsymbol{q}_h\|_*$. The first one exhibits superconvergence, and the last two follow optimal convergence rates, as expected. In addition to the errors, we display the curves associated with both estimators, namely η and $\widetilde{\eta} := \|\boldsymbol{\nabla}\varepsilon_h\|_{\mathcal{T}_h}$. Figure 1(d) compares the convergence rates between the L^2-error associated with the mixed BDM scheme and the L^2-error of the postprocessed solution.

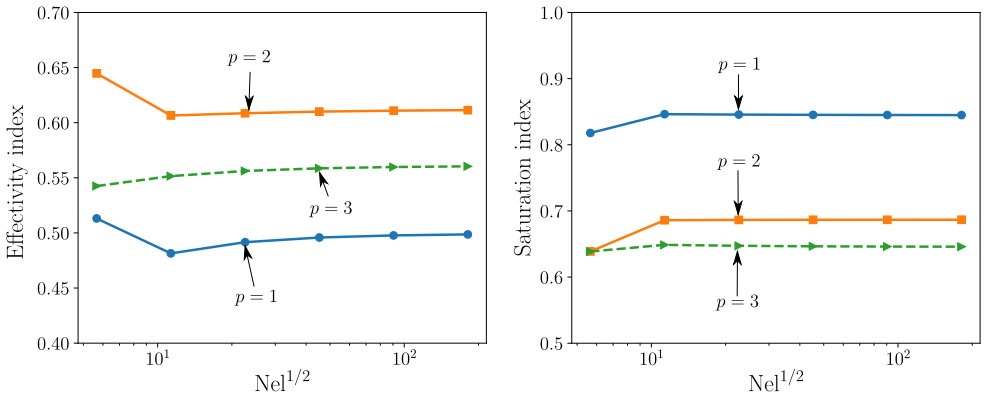
We notice that the behavior is consistent with Remark 2.6; that is, we gain two orders of convergence for any integer $p \geq 2$, and only one order for $p=1$, since $f \notin \mathcal{V}_h^{p-1}$ in this case.

Figure 2(a) shows how the effectivity index of η becomes uniform with respect to h for each polynomial order p under consideration, while Figure 2(b) displays the constant δ involved in the saturation assumption (Assumption 3.5). Notice that the value of δ remains bounded uniformly with respect to h for each polynomial order p. Observe that δ decreases as p increases, which would allow us to conjecture two things: $\delta < 1$ for all $p \in \mathbb{N}$; and, as p increases, a smaller reliability constant in (3.4) is obtained (see the proof of Theorem 3.6).

4.2. A low-regularity solution. We consider again the Poisson problem (2.4) but now on the L-shaped domain $\Omega := (-1,1)^2 \setminus (-1,0]^2$. We set $f \equiv 0$ and u_D such that the exact solution in polar coordinates is $u(r,\vartheta) = r^{2/3}\sin\left(\frac{2}{3}(\pi-\vartheta)\right)$. We

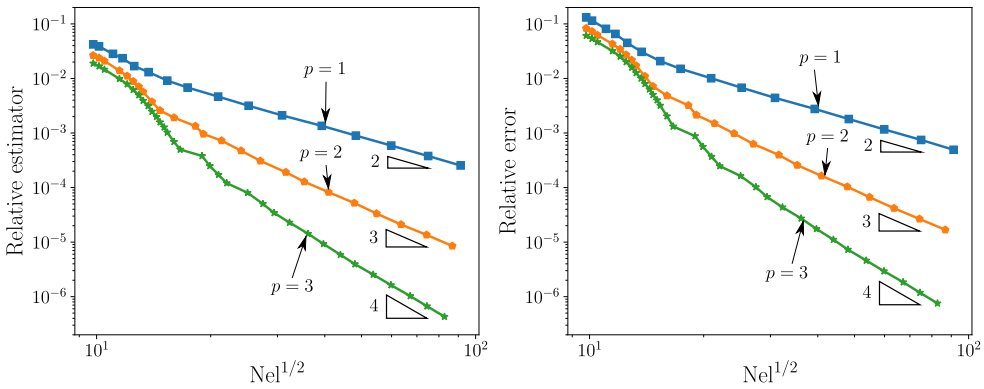
Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.



(a) Effectivity index of η vs. Nel$^{1/2}$ (b) Saturation constant δ vs. Nel$^{1/2}$

FIG. 2. *Effectivity index and saturation constant for a smooth solution scenario.*



(a) η vs. Nel$^{1/2}$ (b) Full error vs. Nel$^{1/2}$

FIG. 3. *Convergence curves for a low-regularity solution scenario.*

note that $u \in H^{5/3-\epsilon}(\Omega)$, for all $\epsilon > 0$, due to a singularity located at the origin $(0,0)$ [42].

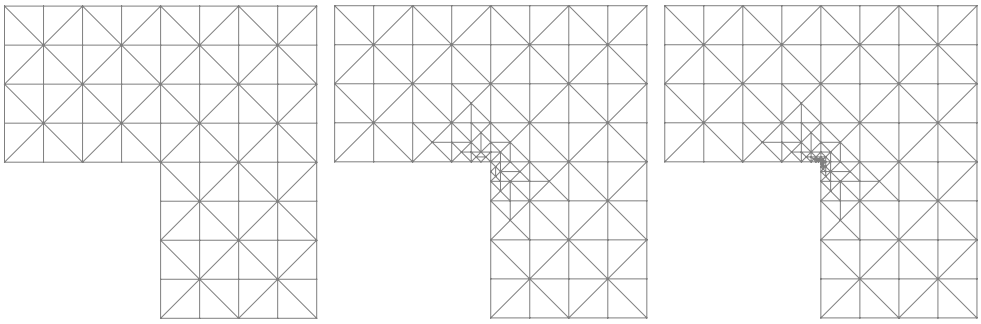
For adaptively refined meshes, Figure 3(a) shows the convergence curves associated with the error estimator η, while Figure 3(b) shows the curves associated with the full error $(\|u - \nu_h\|^2_{1,h} + \|\boldsymbol{q} - \boldsymbol{q}_h\|^2_\Omega)^{1/2}$. Notice that we recover the convergence rates that we would have in a smooth scenario, together with superconvergence.

Furthermore, Figures 4(a), 4(b), and 4(c) display the evolution of the adapted mesh refinement (driven by our estimator η), and how the mesh is locally refined in a neighborhood of the corner $(0,0)$, where the gradient of u has a singularity.

Finally, in Figure 5, we show the L^2-errors of the original finite element (BDM) solution u_h and the residual minimization postprocessed solution ν_h. Notice the increment in the convergence rates when we consider ν_h instead of u_h. The curves show optimal convergence rates for u_h and superconvergence for ν_h, thanks to the use of the adaptive mesh refinement process driven by η. In contrast to Figure 1(d), we gain two orders of convergence for any $p \geq 1$ since $f \equiv 0 \in \mathcal{V}_h^{p-1}$ in this experiment.

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy



(a) Initial (Nel = 96) (b) 5th iteration (Nel = 156) (c) 10th iteration (Nel = 207)

FIG. 4. *Sequence of adapted meshes for a low-regularity solution scenario, with* $p=3$.

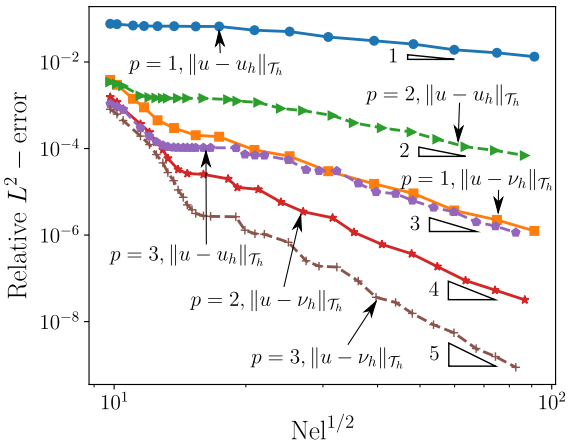


FIG. 5. *Increment in order of convergence.*

4.3. Application to advection-diffusion problems. Let us consider the following advection-diffusion problem in primal and mixed forms: Given $f:\Omega\to\mathbb{R}$, $\boldsymbol{\beta}:\Omega\to\mathbb{R}^2$, $\alpha:\Omega\to\mathbb{R}$, and $u_D:\partial\Omega\to\mathbb{R}$, find $u:\Omega\to\mathbb{R}$ such that

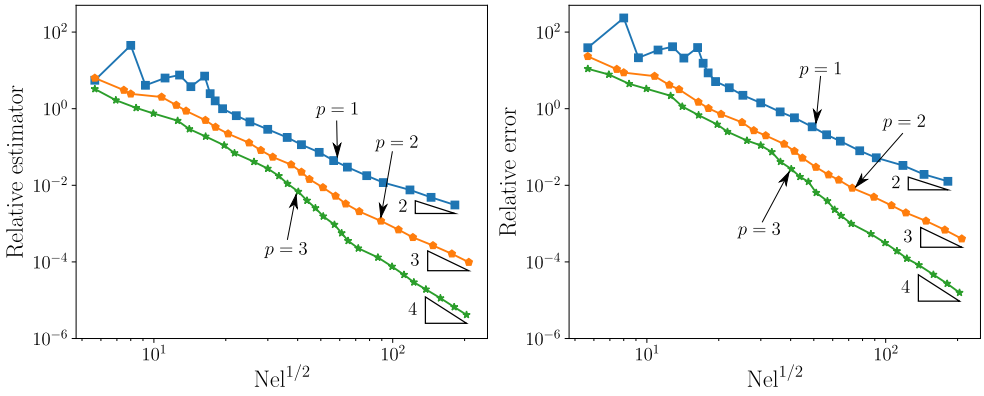
(4.1)
$$\left\{\begin{aligned} -\Delta u+(\boldsymbol{\beta}\cdot\boldsymbol{\nabla})u+\alpha u &= f && \text{in } \Omega,\\ u &= u_D && \text{on } \partial\Omega,\end{aligned}\right. \iff \left\{\begin{aligned} \boldsymbol{q}+\boldsymbol{\nabla}u &= \mathbf{0} && \text{in } \Omega,\\ \boldsymbol{\nabla}\cdot\boldsymbol{q}-\boldsymbol{\beta}\cdot\boldsymbol{q}+\alpha u &= f && \text{in } \Omega,\\ u &= u_D && \text{on } \partial\Omega,\end{aligned}\right.$$

We set $\mathrm{P}=10^3/3$, $\boldsymbol{\beta}\equiv(\mathrm{P},\mathrm{P})$, and $\alpha\equiv 0$. We consider f and u_D such that the exact solution to (4.1) is $u(x_1,x_2)=(x_1+(e^{\mathrm{P}x_1}-1)/(1-e^{\mathrm{P}}))(x_2+(e^{\mathrm{P}x_2}-1)/(1-e^{\mathrm{P}}))$.

We approximate the solution to (4.1) through a BDM finite element discretization. Then, we solve the residual minimization problem (3.1) to postprocess the discrete solution and extract $\|\boldsymbol{\nabla}\varepsilon_h\|_{\mathcal{T}_h}$ to construct the improved estimator η (used to drive adaptive mesh refinements). Figure 6(a) displays the convergence curves of the error estimator η, while Figure 6(b) displays the convergence curves of the full error $(\|u-\nu_h\|_{1,h}^2+\|\boldsymbol{q}-\boldsymbol{q}_h\|_\Omega^2)^{1/2}$. Despite the dominant convection regime, which usually impairs convergence, we have obtained optimal convergence rates.

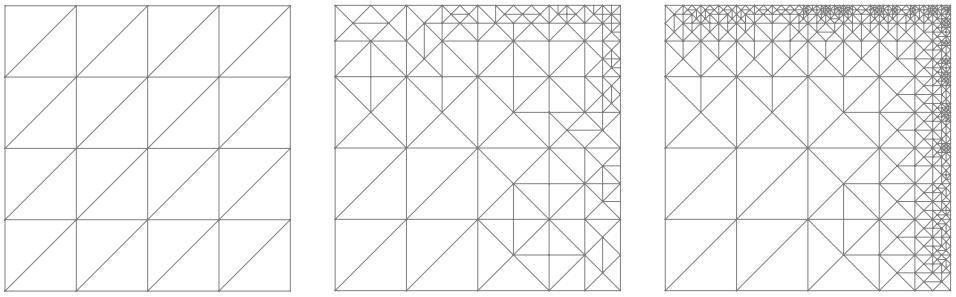
Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy



(a) η vs. Nel$^{1/2}$ (b) Full error vs. Nel$^{1/2}$

FIG. 6. *Convergence curves for the advection-diffusion problem.*



(a) Initial (Nel = 32) (b) 5th iteration (Nel = 196) (c) 10th iteration (Nel = 887)

FIG. 7. *Sequence of adapted meshes, advection-diffusion problem scenario, with* $p=3$.

In addition, Figure 7 shows how our method delivers adaptive meshes that sharply capture the boundary layer, as we can observe from highly localized refinements around the outflow boundary.

Finally, in Figure 8, we see the contrast between the behavior of the L^2-error associated with the BDM mixed method and the postprocessing scheme. We observe the improvement in the rates of convergence for each polynomial approximation degree. In particular, notice the lack of convergence of the BDM mixed method for $p=1$ and $p=2$. which is optimally restored by our scheme. As in the smooth solution experiment (see Figure 1(d)), we have that $f \notin \mathcal{V}_h^{p-1}$, so the first estimate in (2.15) does not hold for $p=1$.

5. Conclusions. Inspired by the postprocessing technique due to Stenberg [46], we have proposed an adaptive FEM based on residual minimization, suitable for PDEs involving a diffusion term. The residual minimization approach is used on the postprocessing scheme associated with a mixed finite element discretization of our model problem. In particular, we have considered the BDM FEM for simplicity. However, the extension to other mixed schemes (such as RT and HDG, to name just a few) and to more complex models (such as advection-diffusion-reaction or Stokes-like problems) is straightforward.

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

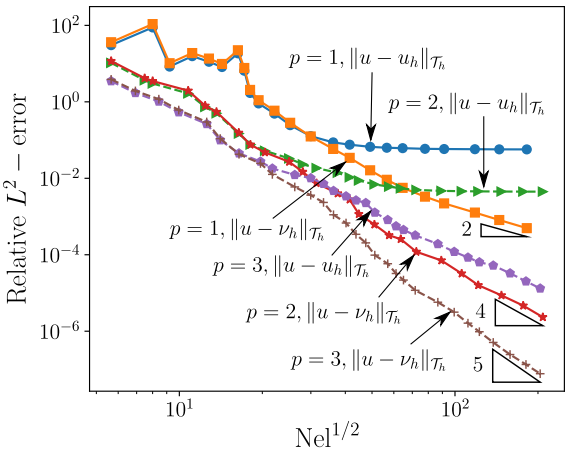


FIG. 8. *Increment in order of convergence.*

Our proposed method has two main advantages. First, the residual minimization approach computes a built-in residual representative, which has proven to be a reliable and efficient a posteriori error estimator in a weak norm. Second, since we perform residual minimization on a local postprocessing scheme, we can solve the underlying saddle point problem with minimal computational effort. Thus, our approach becomes attractive for problems involving different kinds of local behaviors needing adaptive mesh refinements. Our scheme provides, in addition to the residual representative, a superconvergent postprocessing of the scalar variable, which coincides with Stenberg's postprocessed solution.

We have also proposed an improved a posteriori error estimator by adding the jump of the postprocessed solution, as well as a residual term quantifying the mismatch between the discrete flux and the gradient of the postprocessed solution. This improved estimator is proven to be reliable and efficient in a stronger norm. Moreover, like the built-in estimator, it can be used in a wide spectrum of problems involving diffusion terms since both estimators are PDE-independent.

Future research challenges include extensions to other families of mixed formulations supporting postprocessing schemes, such as, for instance, HDG methods.

Appendix A. Auxiliary estimates. We start with an inequality for polynomials lying in the complementary subspace of the kernel of the restricted trace operator $\gamma_K|_{\mathcal{V}^{p+2}_{\star,K}}$, for each $K \in \mathcal{T}_h$.

LEMMA A.1. *Given $K \in \mathcal{T}_h$, let $\mathcal{V}_{\partial K} \subset \mathcal{V}^{p+2}_{\star,K}$ be the complementary subspace of* $\ker\big(\gamma_K|_{\mathcal{V}^{p+2}_{\star,K}}\big)$. *Then,*

$$\|\boldsymbol{\nabla} v_K\|_K \lesssim h_K^{-1/2}\|v_K\|_{\partial K} \qquad \forall v_K \in \mathcal{V}_{\partial K}. \tag{A.1}$$

Proof. Let $\widehat{K}$ be the reference simplex. Since $\|\cdot\|_{\partial\widehat{K}}$ and $\|\boldsymbol{\nabla}(\cdot)\|_{\widehat{K}}$ are norms in the finite-dimensional vector space $\mathcal{V}_{\partial\widehat{K}} \subset \mathcal{V}^{p+2}_{\star,\widehat{K}}$, they are equivalent norms. Then, we have $\|\boldsymbol{\nabla} v_{\widehat{K}}\|_{\widehat{K}} \lesssim \|v_{\widehat{K}}\|_{\partial\widehat{K}}$. The result follows from a standard scaling argument (see [29, Lemma 1.101]). ∎

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

LEMMA A.2. *Let $(u,\boldsymbol{q})\in\mathcal{V}\times\boldsymbol{\mathcal{H}}$ solve the continuous problem (2.5); let $(u_h,\boldsymbol{q}_h)\in\mathcal{V}_h^{p-1}\times\boldsymbol{\mathcal{H}}_h^p$ solve the discrete problem (2.6); let $\theta_h\in\mathcal{V}_{\star,h}^{p+2}$ solve (3.3); and let $(\varepsilon_h,\nu_h)\in\mathcal{V}_{\star,h}^{p+2}\times\mathcal{V}_h^{p+1}$ solve (3.2). Then, for all $K\in\mathcal{T}_h$, the following holds true:*

$$\|\boldsymbol{\nabla}(\theta_h-\nu_h)\|_K=\widetilde{\eta}_K, \tag{A.2}$$

$$\|\boldsymbol{q}-\boldsymbol{q}_h\|_{*,K}\le\widetilde{\eta}_K+\|\boldsymbol{\nabla}(u-\nu_h)\|_K. \tag{A.3}$$

Proof. Using the first equation of (3.2), we get

$$\begin{aligned}(\boldsymbol{\nabla}(\theta_h-\nu_h),\boldsymbol{\nabla}v_K)_K&=-(\boldsymbol{\nabla}\nu_h,\boldsymbol{\nabla}v_K)_K+(\boldsymbol{\nabla}\theta_h,\boldsymbol{\nabla}v_K)_K\\&=(\boldsymbol{\nabla}\varepsilon_h,\boldsymbol{\nabla}v_K)_K+(\boldsymbol{q}_h,\boldsymbol{\nabla}v_K)_K+(\boldsymbol{\nabla}\theta_h,\boldsymbol{\nabla}v_K)_K\\&=(\boldsymbol{\nabla}\varepsilon_h,\boldsymbol{\nabla}v_K)_K\end{aligned}$$

for all $v_K\in\mathcal{V}_{\star,K}^{p+2}$. Thus, we have

$$\begin{aligned}\|\boldsymbol{\nabla}(\theta_h-\nu_h)\|_K&=\sup_{v_K\in\mathcal{V}_{\star,K}^{p+2}\setminus\{0\}}\frac{(\boldsymbol{\nabla}(\theta_h-\nu_h),\boldsymbol{\nabla}v_K)_K}{\|\boldsymbol{\nabla}v_K\|_K}=\sup_{v_K\in\mathcal{V}_{\star,K}^{p+2}\setminus\{0\}}\frac{(\boldsymbol{\nabla}\varepsilon_h,\boldsymbol{\nabla}v_K)_K}{\|\boldsymbol{\nabla}v_K\|_K}\\&=\|\boldsymbol{\nabla}\varepsilon_K\|_K=\widetilde{\eta}_K,\end{aligned}$$

which proves (A.2). Now, we deal with the flux error. For each $K\in\mathcal{T}_h$, we notice that $(\boldsymbol{q}-\boldsymbol{q}_h,\boldsymbol{\nabla}v_K)_K=(\boldsymbol{\nabla}\varepsilon_h,\boldsymbol{\nabla}v_K)_K-(\boldsymbol{\nabla}(u-\nu_h),\boldsymbol{\nabla}v_K)_K$ for all $v_K\in\mathcal{V}_{\star,K}^{p+2}$, and recalling (2.2), we have

$$\|\boldsymbol{q}-\boldsymbol{q}_h\|_{*,K}\le\|\boldsymbol{\nabla}\varepsilon_h\|_K+\|\boldsymbol{\nabla}(u-\nu_h)\|_K=\widetilde{\eta}_K+\|\boldsymbol{\nabla}(u-\nu_h)\|_K$$

for all $K\in\mathcal{T}_h$, which proves (A.3). ∎

Appendix B. Controlling $h_K\|\boldsymbol{\nabla}\cdot(\boldsymbol{q}-\boldsymbol{q}_h)\|_K+h_K^{1/2}\|(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}\|_{\partial K}$. We first establish a general (Fortin) assumption that will be verified in Appendix C for the lowest-order case $p=1$.

Assumption B.1 (Fortin operator). Let $\gamma_K:H^1(K)\to L^2(\partial K)$ denote the continuous Dirichlet trace operator. There exist an operator $\Pi_{\partial K}:L^2(\partial K)\to\gamma_K(\mathcal{V}_{\star,K}^{p+2})$ and an h-independent constant $C_\Pi>0$, such that

$$\|\Pi_{\partial K}v_K\|_{\partial K}\le C_\Pi\|v_K\|_{\partial K}\qquad\forall v_K\in L^2(\partial K), \tag{B.1a}$$

$$(\boldsymbol{p}_h\cdot\boldsymbol{n},v_K-\Pi_{\partial K}v_K)_{\partial K}=0\quad\forall\boldsymbol{p}_h\in\boldsymbol{\mathcal{H}}_h^p,\ \forall v_K\in L^2(\partial K). \tag{B.1b}$$

PROPOSITION B.2. *Let $(u,\boldsymbol{q})\in\mathcal{V}\times\boldsymbol{\mathcal{H}}$ be the solution of the continuous variational formulation associated with (3.7), following (2.5), and assume $u\in H^s(\Omega)$ with $s>3/2$; let $(u_h,\boldsymbol{q}_h)\in\mathcal{V}_h^{p-1}\times\boldsymbol{\mathcal{H}}_h^p$ be the BDM finite element approximation of (3.7); and let $(\varepsilon_h,\nu_h)\in\mathcal{V}_{\star,h}^{p+2}\times\mathcal{V}_h^{p+1}$ solve (3.2) for all $K\in\mathcal{T}_h$. Then, for each $K\in\mathcal{T}_h$, we have*

$$\begin{aligned}&h_K\|\boldsymbol{\nabla}\cdot(\boldsymbol{q}-\boldsymbol{q}_h)\|_K+h_K^{1/2}\|(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}\|_{\partial K}\\&\qquad\lesssim\eta_K+h_K\|f-\boldsymbol{\nabla}\cdot\boldsymbol{q}_h-\mathcal{R}\nu_h\|_K+h_K\|\mathcal{R}(u-\nu_h)\|_K+\operatorname{osc}_K(\boldsymbol{q}),\end{aligned} \tag{B.2}$$

where $\operatorname{osc}_K(\boldsymbol{q}):=h_K^{1/2}\sup_{v_K\in L^2(\partial K)\setminus\{0\}}\frac{\langle\boldsymbol{q}\cdot\boldsymbol{n},v_K-\Pi_{\partial K}v_K\rangle_{\partial K}}{\|v_K\|_{\partial K}}$.

Proof. We notice that, for each $K\in\mathcal{T}_h$,

$$\|(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}\|_{\partial K}=\sup_{v_K\in L^2(\partial K)\setminus\{0\}}\frac{\langle(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n},v_K\rangle_{\partial K}}{\|v_K\|_{\partial K}}.$$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Thus, thanks to (B.1) and Lemma A.1,[7] we get

$$\begin{aligned}
\|(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}\|_{\partial K} &\le \sup_{v_K\in L^2(\partial K)\setminus\{0\}} \frac{\langle(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}, v_K-\Pi_{\partial K}v_K\rangle_{\partial K}}{\|v_K\|_{\partial K}} \\
&\quad + \sup_{v_K\in L^2(\partial K)\setminus\{0\}} \frac{\langle(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}, \Pi_{\partial K}v_K\rangle_{\partial K}}{\|v_K\|_{\partial K}} \\
&\le \sup_{v_K\in L^2(\partial K)\setminus\{0\}} \frac{\langle(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}, v_K-\Pi_{\partial K}v_K\rangle_{\partial K}}{\|v_K\|_{\partial K}} \\
&\quad + C_\Pi \left(\sup_{v_K\in L^2(\partial K)\setminus\{0\}} \frac{\langle(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}, \Pi_{\partial K}v_K\rangle_{\partial K}}{\|\Pi_{\partial K}v_K\|_{\partial K}} \right) \\
&\le h_K^{-1/2}\mathrm{osc}_K(\boldsymbol{q}) + C_\Pi \left(\sup_{w_K\in\gamma_K\left(\mathcal{V}_{\star,K}^{p+2}\right)\setminus\{0\}} \frac{\langle(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}, w_K\rangle_{\partial K}}{\|w_K\|_{\partial K}} \right) \\
&= h_K^{-1/2}\mathrm{osc}_K(\boldsymbol{q}) + C_\Pi \left(\sup_{w_K\in\gamma_K\left(\mathcal{V}_{\partial K}\right)\setminus\{0\}} \frac{\langle(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}, w_K\rangle_{\partial K}}{\|w_K\|_{\partial K}} \right) \\
&\le h_K^{-1/2}\mathrm{osc}_K(\boldsymbol{q}) + C_\Pi \left(\sup_{w_K\in\gamma_K\left(\mathcal{V}_{\partial K}\right)\setminus\{0\}} \frac{\langle(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}, w_K\rangle_{\partial K}}{h_K^{1/2}\|\boldsymbol{\nabla} w_K\|_K} \right).
\end{aligned}$$

Therefore, we conclude (B.2) by noticing that

$$\begin{aligned}
\langle(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}, w_K\rangle_{\partial K} &= (\boldsymbol{\nabla}\varepsilon_K, \boldsymbol{\nabla} w_K)_K - (\boldsymbol{\nabla}(u-\nu_h), \boldsymbol{\nabla} w_K)_K + (\boldsymbol{\nabla}\cdot(\boldsymbol{q}-\boldsymbol{q}_h), w_K)_K \\
&\le \left(\widetilde{\eta}_K + \|\boldsymbol{\nabla}(u-\nu_h)\|_K + h_K\|\boldsymbol{\nabla}\cdot(\boldsymbol{q}-\boldsymbol{q}_h)\|_K\right)\|\boldsymbol{\nabla} w_K\|_K,
\end{aligned}$$

as well as $\widetilde{\eta}_K + \|\boldsymbol{\nabla}(u-\nu_h)\|_K \lesssim \eta_K$ and

$$h_K\|\boldsymbol{\nabla}\cdot(\boldsymbol{q}-\boldsymbol{q}_h)\|_K \le h_K\|f-\boldsymbol{\nabla}\cdot\boldsymbol{q}_h-\mathcal{R}\nu_h\|_K + h_K\|\mathcal{R}(u-\nu_h)\|_K. \qquad \square$$

To close this appendix, we state a posteriori error estimates for a stronger norm of the error, whose proof is a consequence of Theorems 3.9 and 3.11 and Proposition B.2.

COROLLARY B.3. *Under the same hypotheses of Proposition* B.2*, let us further assume that* $\mathcal{R}$ *(see Remark* 3.7*) satisfies*[8]

$$\sum_{K\in\mathcal{T}_h} h_K^2\|\mathcal{R}(u-\nu_h)\|_K^2 \lesssim \eta^2.$$

Then, we have the following reliability estimate:

$$\|u-\nu_h\|_{1,h}^2 + \|\boldsymbol{q}-\boldsymbol{q}_h\|_{\mathrm{div},h}^2 \lesssim \eta^2 + \sum_{K\in\mathcal{T}_h} h_K^2\|f-\boldsymbol{\nabla}\cdot\boldsymbol{q}_h-\mathcal{R}\nu_h\|_K^2, \tag{B.3}$$

and, for all $K\in\mathcal{T}_h$*, the following efficiency estimate holds:*

$$\eta_K^2 + h_K^2\|f-\boldsymbol{\nabla}\cdot\boldsymbol{q}_h-\mathcal{R}\nu_h\|_K^2 \lesssim |u-\nu_h|_{1,K,h}^2 + \|\boldsymbol{q}-\boldsymbol{q}_h\|_{\mathrm{div},K,h}^2, \tag{B.4}$$

where $\|\boldsymbol{q}-\boldsymbol{q}_h\|_{\mathrm{div},K,h}^2 := \|\boldsymbol{q}-\boldsymbol{q}_h\|_K^2 + h_K^2\|\boldsymbol{\nabla}\cdot(\boldsymbol{q}-\boldsymbol{q}_h)\|_K^2 + h_K\|(\boldsymbol{q}-\boldsymbol{q}_h)\cdot\boldsymbol{n}\|_{\partial K}^2$ *and* $\|\boldsymbol{q}-\boldsymbol{q}_h\|_{\mathrm{div},h}^2 := \sum_{K\in\mathcal{T}_h}\|\boldsymbol{q}-\boldsymbol{q}_h\|_{\mathrm{div},K,h}^2$.

[7] Notice that the range of $\gamma_K|_{\mathcal{V}_{\star,K}^{p+2}}$ equals the range of $\gamma_K|_{\mathcal{V}_{\partial K}}$.

[8] For instance, the operator $\mathcal{R}$ associated with the advection-diffusion-reaction equation (4.1) satisfies $\sum_{K\in\mathcal{T}_h} h_K^2\|\mathcal{R}(u-\nu_h)\|_K^2 \lesssim \sum_{K\in\mathcal{T}_h} h_K^2(\|\boldsymbol{\beta}\|_{\infty,K}^2\|\boldsymbol{\nabla}(u-\nu_h)\|_K^2 + \|\alpha\|_{\infty,K}^2\|u-\nu_h\|_K^2) \lesssim \eta^2$, thanks to Theorem 3.9 and Remark 3.10.

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Appendix C. Verification of Assumption B.1. For our choice of a mixed FEM (i.e., BDM), we verify Assumption B.1 for the lowest-order finite element spaces $\mathcal{V}^3_{\star,h}$ and $\boldsymbol{\mathcal{H}}^1_h$ in dimension $d=2$. We start by stating the following auxiliary lemma.

LEMMA C.1. *For each $K \in \mathcal{T}_h$, let $\{\varphi_i^K\}_{i=1}^6$ be the set of nodal basis functions spanning $\{\mu \in L^2(\partial K) : \exists \boldsymbol{z} \in \boldsymbol{\mathcal{H}}^1_h$ such that $\mu = \boldsymbol{z}|_K \cdot \boldsymbol{n}_K\}$. There exists a (biorthogonal) set $\{\psi_i^K\}_{i=1}^6 \subset \mathcal{V}^3_{\star,K}$ such that*

$$\int_{\partial K} \varphi_i^K \psi_j^K = \xi_K \delta_{ij}, \qquad i,j = 1,\dots,6\,, \tag{C.1}$$

where δ_{ij} denotes the Kronecker delta, and $\xi_K\,(\approx h_K)$ denotes a scaling constant of the element K. Moreover, for all $K \in \mathcal{T}_h$, we have the estimate

$$\|\psi_i^K\|_{\partial K} \lesssim \xi_K^{1/2}, \qquad i = 1,\dots,6\,. \tag{C.2}$$

Proof. Consider the reference simplex $\widehat{K} := \{(s,t) \in \mathbb{R}^2 : s \in [0,1],\, t \in [0, 1-s]\}$,, and let $\{\widehat{\varphi}_i^{\widehat{K}}\}_{i=1}^6$ be defined by $\widehat{\varphi}_i^{\widehat{K}} = \varphi_i^K \circ \mathcal{F}_K$, where $\mathcal{F}_K : \widehat{K} \to K$ is the affine mapping satisfying $K = \mathcal{F}_K(\widehat{K})$. We will construct $\{\widehat{\psi}_i^{\widehat{K}}\}_{i=1}^6$ such that

$$\int_{\partial \widehat{K}} \widehat{\varphi}_i^{\widehat{K}} \widehat{\psi}_j^{\widehat{K}} = \delta_{ij}, \qquad i,j = 1,\dots,6, \tag{C.3}$$

and thus we will define $\psi_i^K := \widehat{\psi}_i^{\widehat{K}} \circ \mathcal{F}_K^{-1}$.

Without loss of generality, let $\widehat{\varphi}_1^{\widehat{K}}$, $\widehat{\varphi}_2^{\widehat{K}}$ be the nodal basis functions associated with the face $\widehat{F}_1 := \{(s,0) \in \mathbb{R}^2 : s \in [0,1]\}$; let $\widehat{\varphi}_3^{\widehat{K}}$, $\widehat{\varphi}_4^{\widehat{K}}$ be the nodal basis functions associated with the face $\widehat{F}_2 := \{(1-s,s) \in \mathbb{R}^2 : s \in [0,1]\}$; and let $\widehat{\varphi}_5^{\widehat{K}}$, $\widehat{\varphi}_6^{\widehat{K}}$ be the nodal basis functions associated with the face $\widehat{F}_3 := \{(0,s) \in \mathbb{R}^2 : s \in [0,1]\}$. We are going to exhibit three functions $\{\widehat{\phi}_j\}_{j=1}^3 \subset \mathcal{V}^3_{*,\widehat{K}}$, such that

$$\widehat{\phi}_j|_{\widehat{F}_k} = 0 \qquad \text{for } j = 1,2,3 \quad \text{and} \quad k = 2,3. \tag{C.4}$$

Indeed, for $\widehat{\lambda}_1(\widehat{x}_1,\widehat{x}_2) := 1 - \widehat{x}_1 - \widehat{x}_2$ and $\widehat{\lambda}_2(\widehat{x}_1,\widehat{x}_2) := \widehat{x}_1$, we set $\widehat{\phi}_1 := \widehat{\lambda}_1\widehat{\lambda}_2$, $\widehat{\phi}_2 := \big(\widehat{\lambda}_1\big)^2\widehat{\lambda}_2$, and $\widehat{\phi}_3 := \widehat{\lambda}_1\big(\widehat{\lambda}_2\big)^2$. We propose to find linear combinations

$$\widehat{\psi}_1^{\widehat{K}} := \beta_1\widehat{\phi}_1 + \beta_2\widehat{\phi}_2 + \beta_3\widehat{\phi}_3 \qquad \text{and} \qquad \widehat{\psi}_2^{\widehat{K}} := \gamma_1\widehat{\phi}_1 + \gamma_2\widehat{\phi}_2 + \gamma_3\widehat{\phi}_3, \tag{C.5}$$

such that

$$\int_{\widehat{F}_1} \widehat{\varphi}_i^{\widehat{K}} \widehat{\psi}_j^{\widehat{K}} = \delta_{ij}, \qquad i,j = 1,2, \qquad \text{and} \qquad \int_{\widehat{K}} \widehat{\psi}_j^{\widehat{K}} = 0, \qquad j = 1,2. \tag{C.6}$$

Notice that (C.4) and (C.6) guarantee (C.3). We can condensate (C.5) and (C.6) as the linear systems $\boldsymbol{A}\boldsymbol{\beta} = \boldsymbol{e}_1$ and $\boldsymbol{A}\boldsymbol{\gamma} = \boldsymbol{e}_2$, where $\boldsymbol{\beta} := (\beta_1,\beta_2,\beta_3)$, $\boldsymbol{\gamma} := (\gamma_1,\gamma_2,\gamma_3)$, $\boldsymbol{e}_1 = (1,0,0)$, $\boldsymbol{e}_2 = (0,1,0)$, and

$$\boldsymbol{A} = \frac{1}{120}\begin{bmatrix} 10 & 6 & 4 \\ 10 & 4 & 6 \\ 5 & 2 & 2 \end{bmatrix}, \qquad \text{where } A_{ij} := \begin{cases} \int_{\widehat{F}_1} \widehat{\varphi}_i^{\widehat{K}}\widehat{\phi}_j\,, & i = 1,2;\ j = 1,2,3, \\ \int_{\widehat{K}} \widehat{\phi}_j\,, & i = 3;\ j = 1,2,3. \end{cases}$$

Since $\boldsymbol{A}$ is invertible, we can determine $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ uniquely, completing the construction in (C.5). Thus, we obtain (C.3) after constructing $\{\widehat{\psi}_3^{\widehat{K}}, \widehat{\psi}_4^{\widehat{K}}\}$ and $\{\widehat{\psi}_5^{\widehat{K}}, \widehat{\psi}_6^{\widehat{K}}\}$

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

with the same arguments. Hence, (C.1) follows from (C.3) by standard scaling arguments with $\xi_K := |\partial K|/|\partial \widehat{K}|$, where $|\cdot|$ denotes the measure (i.e., perimeter in this particular case). Finally (see [29, Lemma 1.101]), (C.2) follows from the fact that $\|\psi_i^K\|_{\partial K} \lesssim \xi_K^{1/2} \|\widehat{\psi}_i^{\widehat{K}}\|_{\partial \widehat{K}} \lesssim \xi_K^{1/2}$, $i = 1, \dots, 6$. ∎

Now, we construct the Fortin operator. Let us consider, for each $K \in \mathcal{T}_h$, the operator $\Pi_{\partial K} : L^2(\partial K) \to \gamma_K(\mathcal{V}_{\star,K}^{p+2})$ defined by $\Pi_{\partial K} v := \sum_{j=1}^6 \alpha_j^K(v)\, \psi_j^K$, where $\alpha_j^K(v) \in \mathbb{R}$ is chosen to satisfy

$$\int_{\partial K} \varphi_i^K \Pi_{\partial K} v = \int_{\partial K} \varphi_i^K v, \qquad i = 1, \dots, 6.$$

Solving for $\alpha_i^K(v)$, and thanks to (C.1), we obtain

$$\alpha_i^K(v) = \frac{1}{\xi_K} \int_{\partial K} \varphi_i^K v, \qquad i = 1, \dots, 6.$$

Moreover, we have

$$|\alpha_i^K(v)| \le \frac{1}{\xi_K} \|\varphi_i^K\|_{\partial K} \|v\|_{\partial K} \le \frac{|\partial K|^{1/2}}{\xi_K} \|v\|_{\partial K}, \qquad i = 1, \dots, 6.$$

Thus, thanks to (C.2) and since $\xi_K = |\partial K|/|\partial \widehat{K}|$, we get

$$\|\Pi_{\partial K} v\|_{\partial K}^2 \lesssim \sum_{j=1}^6 |\alpha_j^K(v)|^2 \|\psi_j^K\|_{\partial K}^2 \lesssim \frac{|\partial K|}{\xi_K} \|v\|_{\partial K}^2 = |\partial \widehat{K}| \|v\|_{\partial K}^2 \lesssim \|v\|_{\partial K}^2.$$

Acknowledgments. The authors thank Carlos Gónzalez-Moraga and José Hasbani for helping with preliminary numerical experiments, Professor David Pardo for his insightful suggestions, and the anonymous reviewers for their careful reviews and comments, leading to an improvement of the original presentation.

REFERENCES

[1] R. Adams and J. Fournier, *Sobolev Spaces*, Academic Press, New York, 2003.
[2] J. H. Adler, T. A. Manteuffel, S. F. McCormick, J. W. Nolting, J. W. Ruge, and L. Tang, *Efficiency based adaptive local refinement for first-order system least-squares formulations*, SIAM J. Sci. Comput., 33 (2011), pp. 1–24, https://doi.org/10.1137/100786897.
[3] M. Ainsworth and X. Ma, *Non-uniform order mixed FEM approximation: Implementation, post-processing, computable error bound and adaptivity*, J. Comput. Phys., 231 (2012), pp. 436–453.
[4] M. Ainsworth and J. T. Oden, *A Posteriori Error Estimation in Finite Element Analysis*, Wiley, New York, 2000.
[5] R. Araya, M. Solano, and P. Vega, *Analysis of an adaptive HDG method for the Brinkman problem*, IMA J. Numer. Anal., 39 (2019), pp. 1502–1528.
[6] R. Araya, M. Solano, and P. Vega, *A posteriori error analysis of an HDG method for the Oseen problem*, Appl. Numer. Math., 146 (2019), pp. 291–308.
[7] D. N. Arnold, *Mixed finite element methods for elliptic problems*, Comput. Methods Appl. Mech. Engrg., 82 (1990), pp. 281–300.
[8] D. N. Arnold and F. Brezzi, *Mixed and nonconforming finite element methods: Implementation, postprocessing and error estimates*, RAIRO Modél. Math. Anal. Numér., 19 (1985), pp. 7–32.
[9] I. Babuška, *Error-bounds for finite element method*, Numer. Math., 16 (1971), pp. 322–333.
[10] R. E. Bank, A. H. Sherman, and A. Weiser, *Some refinement algorithms and data structures for regular local mesh refinement*, in Scientific Computing: Applications of Mathematics and Computing to the Physical Sciences, Vol. 1, R. Stepleman, et al., eds., IMACS, North-Holland, Amsterdam, 1983, pp. 3–17.

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

[11] L. Beirão da Veiga, *A residual based error estimator for the mimetic finite difference method*, Numer. Math., 108 (2008), pp. 387–406.
[12] M. Berndt, T. A. Manteuffel, and S. F. McCormick, *Local error estimates and adaptive refinement for first-order system least squares (FOSLS)*, Electron. Trans. Numer. Anal., 6 (1997), pp. 35–43.
[13] D. Boffi, F. Brezzi, and M. Fortin, *Mixed Finite Element Methods and Applications*, Springer, Heidelberg, 2013.
[14] J. H. Bramble and J. Xu, *A local post-processing technique for improving the accuracy in mixed finite-element approximations*, SIAM J. Numer. Anal., 26 (1989), pp. 1267–1275, https://doi.org/10.1137/0726073.
[15] F. Brezzi, *On the existence, uniqueness and approximation of saddle-point problems arising from Lagrangian multipliers,* Rev. Française Automat. Informat. Recherche Opérationnelle Sér. Rouge, 8 (1974), pp. 129–151.
[16] F. Brezzi, J. Douglas. Jr., R. Durán, and M. Fortin, *Mixed finite elements for second order elliptic problems in three variables*, Numer. Math., 52 (1987), pp. 237–250.
[17] F. Brezzi, J. Douglas, Jr., and L. D. Marini, *Two families of mixed finite elements for second order elliptic problems*, Numer. Math., 47 (1985), pp. 217–235.
[18] V. M. Calo, A. Ern, I. Muga, and S. Rojas, *An adaptive stabilized conforming finite element method via residual minimization on dual discontinuous Galerkin norms*, Comput. Methods Appl. Mech. Engrg., 363 (2020), 112891.
[19] C. Carstensen, L. Demkowicz, and J. Gopalakrishnan, *A posteriori error control for DPG methods*, SIAM J. Numer. Anal., 52 (2014), pp. 1335–1353, https://doi.org/10.1137/130924913.
[20] P. G. Ciarlet, *The Finite Element Method for Elliptic Problems*, Classics Appl. Math. 40, SIAM, Philadelphia, 2002, https://doi.org/10.1137/1.9780898719208.
[21] B. Cockburn, J. Guzmán, and H. Wang, *Superconvergent discontinuous Galerkin methods for second-order elliptic problems*, Math. Comp., 78 (2009), pp. 1–24.
[22] B. Cockburn and W. Zhang, *A posteriori error estimates for HDG methods*, J. Sci. Comput., 51 (2012), pp. 582–607.
[23] B. Cockburn and W. Zhang, *An a posteriori error estimate for the variable-degree Raviart-Thomas method*, Math. Comp., 83 (2014), pp. 1063–1082.
[24] A. Cohen, W. Dahmen, and G. Welper, *Adaptivity and variational stabilization for convection-diffusion equations*, ESAIM Math. Model. Numer. Anal., 46 (2012), pp. 1247–1273.
[25] L. Demkowicz, *Computing with hp-Adaptive Finite Elements*, Vol. 1, Chapman & Hall/CRC, Boca Raton, FL, 2007.
[26] L. Demkowicz, J. Gopalakrishnan, and A. H. Niemi, *A class of discontinuous Petrov-Galerkin methods. Part* III*: Adaptivity*, Appl. Numer. Math., 62 (2012), pp. 396–427.
[27] D. A. Di Pietro and A. Ern, *Mathematical Aspects of Discontinuous Galerkin Methods*, Vol. 69, Springer Science, 2012.
[28] W. Dörfler, *A convergent adaptive algorithm for Poisson's equation*, SIAM J. Numer. Anal., 33 (1996), pp. 1106–1124.
[29] A. Ern and J.-L. Guermond, *Theory and Practice of Finite Elements*, Appl. Math. Sci. 159, Springer-Verlag, New York, 2004.
[30] A. Ern and J.-L. Guermond, *Finite Elements* II*. Galerkin Approximation, Elliptic and Mixed PDEs*, Springer, Cham, 2021.
[31] T. Führer, *Superconvergent DPG methods for second-order elliptic problems*, Comput. Methods Appl. Math., 19 (2019), pp. 483–502.
[32] M. Giacomini, R. Sevilla, and A. Huerta, *HDGlab: An open-source implementation of the hybridisable discontinuous Galerkin method in MATLAB*, Arch. Comput. Methods Eng., 28 (2021), pp. 1941–1986.
[33] G. Giorgiani, S. Fernández-Méndez, and A. Huerta, *Hybridizable discontinuous Galerkin p-adaptivity for wave propagation problems*, Internat. J. Numer. Methods Fluids, 72 (2013), pp. 1244–1262.
[34] J. Huang and X. Huang, *A hybridizable discontinuous Galerkin method for Kirchhoff plates*, J. Sci. Comput., 78 (2019), pp. 290–320.
[35] Y. Hyon and D. Y. Kwak, *Superconvergence of new mixed finite element spaces*, J. Comput. Appl. Math., 235 (2011), pp. 4265–4271.
[36] O. A. Karakashian and F. Pascal, *Convergence of adaptive discontinuous Galerkin approximations of second-order elliptic problems*, SIAM J. Numer. Anal., 45 (2007), pp. 641–665, https://doi.org/10.1137/05063979X.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.

[37] M. G. Larson and A. Målqvist, *A posteriori error estimates for mixed finite element approximations of elliptic problems*, Numer. Math., 108 (2008), pp. 487–500.
[38] A. Logg, K.-A. Mardal, and G. N. Wells, eds., *Automated Solution of Differential Equations by the Finite Element Method: The FEniCS Book*, Lect. Notes Comput. Sci. Eng. 84, Springer, Heidelberg, 2012.
[39] A. Logg, G. N. Wells, and J. Hake, *DOLFIN: A C++/Python finite element library*, in Automated Solution of Differential Equations by the Finite Element Method: The FEniCS Book, A. Logg, K.-A. Mardal, and G. Wells, eds., Springer, Berlin, Heidelberg, 2012, pp. 173–225.
[40] C. Lovadina and R. Stenberg, *Energy norm a posteriori error estimates for mixed finite element methods*, Math. Comp., 75 (2006), pp. 1659–1674.
[41] J.-C. Nédélec, *Mixed finite elements in* $\mathbb{R}^3$, Numer. Math., 35 (1980), pp. 315–341.
[42] J. T. Oden and A. Patra, *A parallel adaptive strategy for hp finite element computations*, Comput. Methods Appl. Mech. Engrg., 121 (1995), pp. 449–470.
[43] L. Payne and H. Weinberger, *An optimal Poincaré inequality for convex domains*, Arch. Ration. Mech. Anal., 5 (1960), pp. 286–292.
[44] P. A. Raviart and J. M. Thomas, *A mixed finite element method for* 2*-nd order elliptic problems*, in Mathematical Aspects of Finite Element Methods, I. Galligani and E. Magenes, eds., Springer, Berlin, Heidelberg, 1977, pp. 292–315.
[45] M. Rüter and R. Stenberg, *Error-controlled adaptive mixed finite element methods for second-order elliptic equations*, Comput. Mech., 42 (2008), pp. 447–456.
[46] R. Stenberg, *Postprocessing schemes for some mixed finite elements*, ESAIM Math. Model. Numer. Anal., 25 (1991), pp. 151–167.
[47] R. Stenberg, *A nonstandard mixed finite element family*, Numer. Math., 115 (2010), pp. 131–139.
[48] R. Verfürth, *A posteriori error estimation and adaptive mesh-refinement techniques*, J. Comput. Appl. Math., 50 (1994), pp. 67–83.

Downloaded 09/14/23 to 128.243.2.33 . Redistribution subject to SIAM license or copyright; see https://epubs.siam.org/terms-privacy

Copyright © by SIAM. Unauthorized reproduction of this article is prohibited.